



Vietnam National University HCMC
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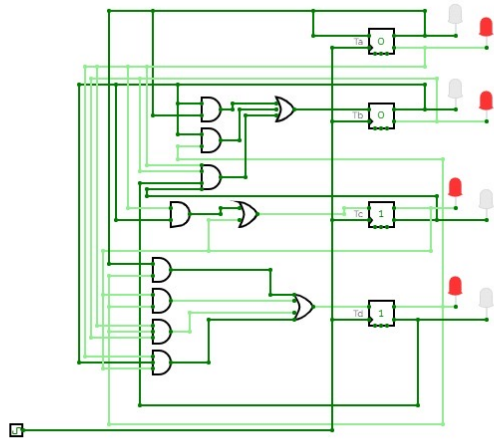


**School of
Electrical Engineering**

EE053IU

Digital Logic Design

Lecture 4: Boolean Algebra and Logic Simplification



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Chapter Objectives

- Apply the basic laws and rules of Boolean algebra
- Apply DeMorgan's theorems to Boolean expressions
- Describe gate combinations with Boolean expressions
- Evaluate Boolean expressions
- Simplify expressions by using the laws and rules of Boolean algebra
- Convert any Boolean expression into a sum-of-products (SOP) form
- Convert any Boolean expression into a product-of-sums (POS) form
- Relate a Boolean expression to a truth table

Chapter Objectives

- Use a Karnaugh map to simplify Boolean expressions
- Use a Karnaugh map to simplify truth table functions
- Utilize “don’t care” conditions to simplify logic functions
- Use the Quine-McCluskey method to simplify Boolean expressions
- Write a VHDL program for simple logic
- Apply Boolean algebra and the Karnaugh map method in an application

1. Boolean Operations and Expressions

- Boolean algebra is the mathematics of digital logic. A basic knowledge of Boolean algebra is indispensable to the study and analysis of logic circuits.
- Variable, complement, and literal are terms used in Boolean algebra.
- A variable is a symbol (usually an italic uppercase letter or word) used to represent an action, a condition, or data. Any single variable can have only a 1 or a 0 value.
- The complement is the inverse of a variable and is indicated by a bar over the variable (overbar).
- A literal is a variable or the complement of a variable.

Boolean Addition

- Boolean addition is equivalent to the OR operation.
- In Boolean algebra, a sum term is a sum of literals.
- In logic circuits, a sum term is produced by an OR operation with no AND operations involved.
- A sum term is equal to 1 when one or more of the literals in the term are 1. A sum term is equal to 0 only if each of the literals is 0.

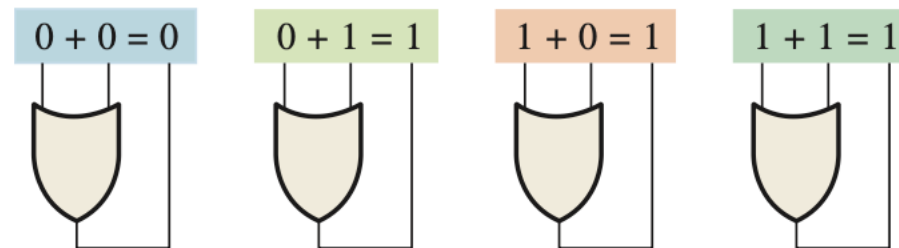


FIGURE 4-1

Example 4–1 Finding Input Values for a Sum Term

1 Problem

Determine the values of A , B , C , and D that make the sum term $A + \bar{B} + C + \bar{D}$ equal to 0.

2 Solution

For the sum term to be 0, each of the literals in the term must be 0. Therefore, $A = 0$, $B = 1$ so that $\bar{B} = 0$, $C = 0$, and $D = 1$ so that $\bar{D} = 0$.

$$A + \bar{B} + C + \bar{D} = 0 + \bar{1} + 0 + \bar{1} = 0 + 0 + 0 + 0 = 0$$

3 Related Problem*

Determine the values of A and B that make the sum term $\bar{A} + B$ equal to 0.

Boolean Multiplication

- Boolean multiplication is equivalent to the AND operation.
- In Boolean algebra, a product term is the product of literals.
- In logic circuits, a product term is produced by an AND operation with no OR operations involved.
- A product term is equal to 1 only if each of the literals in the term is 1. A product term is equal to 0 when one or more of the literals are 0.

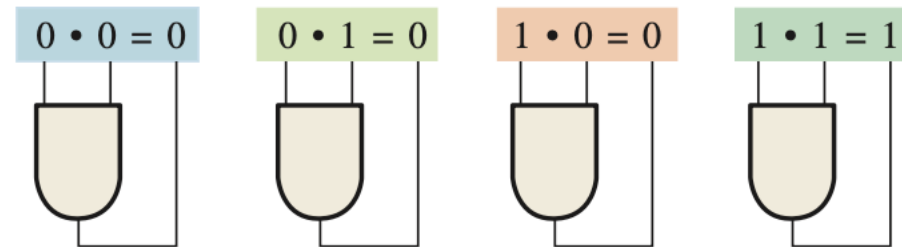


FIGURE 4-2

Boolean Multiplication

Example 4–2 Finding Input Values for a Product Term

1 Problem

Determine the values of A , B , C , and D that make the product term $\overline{A}\overline{B}C\overline{D}$ equal to 1.

2 Solution

For the product term to be 1, each of the literals in the term must be 1.

Therefore, $A = 1$, $B = 0$ so that $\overline{B} = 1$, $C = 1$, and $D = 0$ so that $\overline{D} = 1$.

$$\overline{A}\overline{B}C\overline{D} = 1 \cdot \overline{0} \cdot 1 \cdot \overline{0} = 1 \cdot 1 \cdot 1 \cdot 1 = 1$$

$$\rightarrow A = 1, \quad B = 0, \quad C = 1, \quad D = 0$$

3 Related Problem

Determine the values of A and B that make the product term $\overline{A}\overline{B}$ equal to 1.

2. Laws and Rules of Boolean Algebra

Laws of Boolean Algebra

- The basic laws of Boolean algebra—the commutative laws for addition and multiplication, the associative laws for addition and multiplication, and the distributive law—are the same as in ordinary algebra.
- Each of the laws is illustrated with two or three variables, but the number of variables is not limited to this.

Commutative laws

- The commutative law of addition for two variables is written as:

$$\mathbf{A + B = B + A}$$

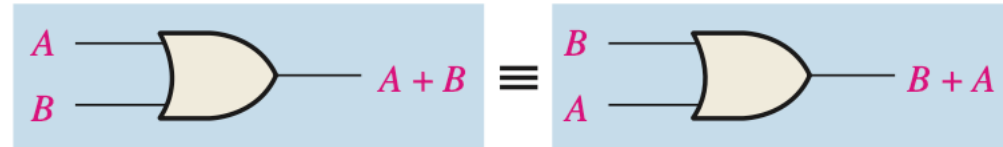


FIGURE 4-3 Application of commutative law of addition.

- The commutative law of multiplication for two variables is written as:

$$\mathbf{AB = BA}$$



FIGURE 4-4 Application of commutative law of multiplication.

Associative laws

- The associative law of addition is written as follows for three variables:

$$\mathbf{A + (B + C) = (A + B) + C}$$

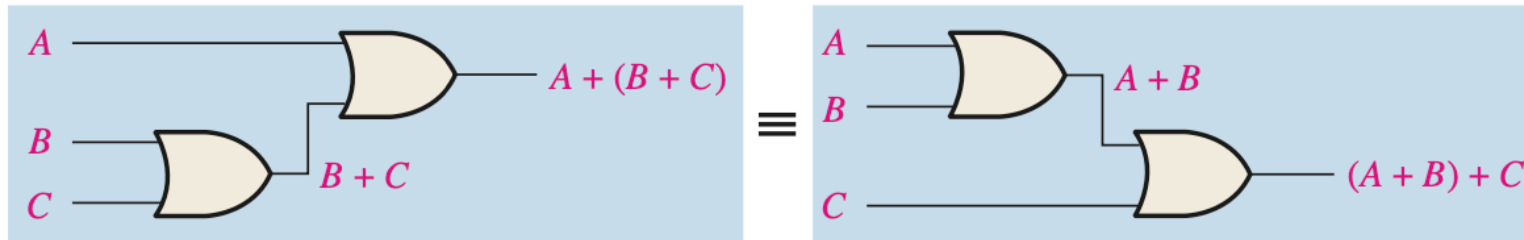


FIGURE 4-5 Application of associative law of addition. Open file F04-05 to verify.

- The associative law of multiplication is written as follows for three variables:

$$\mathbf{A(BC) = (AB)C}$$

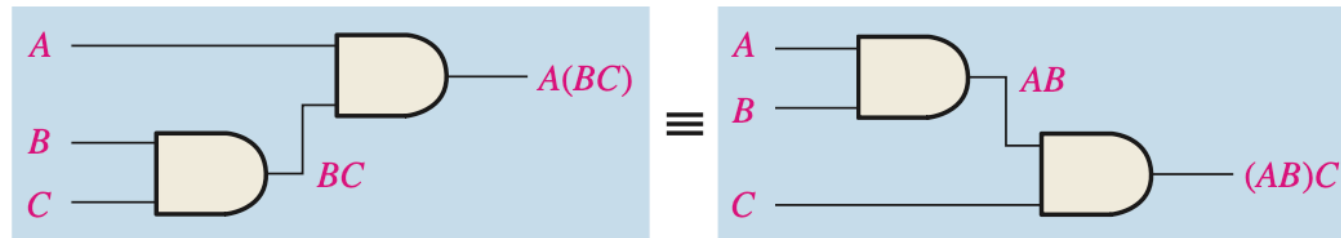


FIGURE 4-6 Application of associative law of multiplication. Open file F04-06 to verify.

Distributive law

- The distributive law is written for three variables as follow:

$$\mathbf{A(B + C) = AB + AC}$$

- This law states that ORing two or more variables and then ANDing the result with a single variable is equivalent to ANDing the single variable with each of the two or more variables and then ORing the products.
- The distributive law also expresses the process of factoring in which the common variable

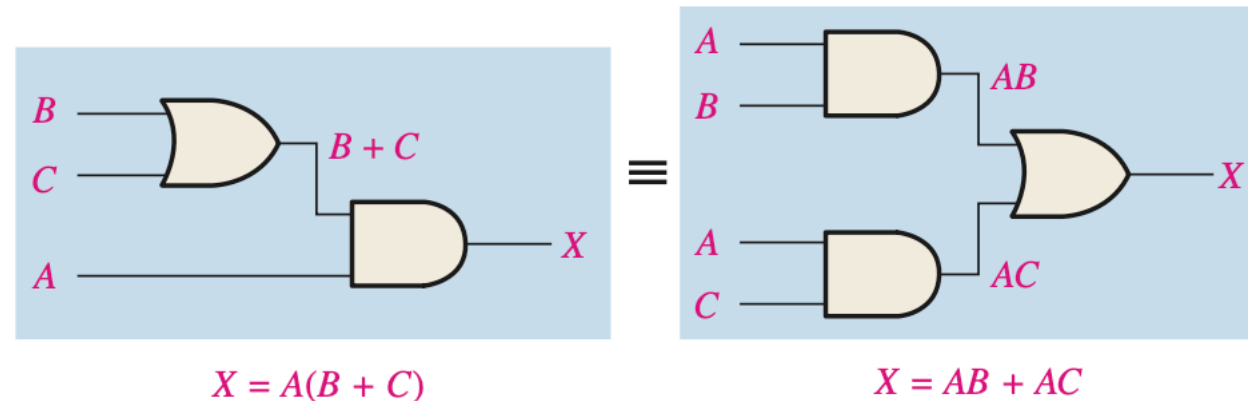


FIGURE 4-7 Application of distributive law. Open file F04-07 to verify.

Rules of Boolean Algebra

TABLE 4-1

Basic rules of Boolean algebra.

1. $A + 0 = A$	7. $A \cdot A = A$
2. $A + 1 = 1$	8. $A \cdot \bar{A} = 0$
3. $A \cdot 0 = 0$	9. $\bar{\bar{A}} = A$
4. $A \cdot 1 = A$	10. $A + AB = A$
5. $A + A = A$	11. $A + \bar{A}B = A + B$
6. $A + \bar{A} = 1$	12. $(A + B)(A + C) = A + BC$

A , B , or C can represent a single variable or a combination of variables.

Rule 1: $A + 0 = A$ A variable ORed with 0 is always equal to the variable. If the input variable A is 1, the output variable X is 1, which is equal to A . If A is 0, the output is 0, which is also equal to A . This rule is illustrated in Figure 4-8, where the lower input is fixed at 0.

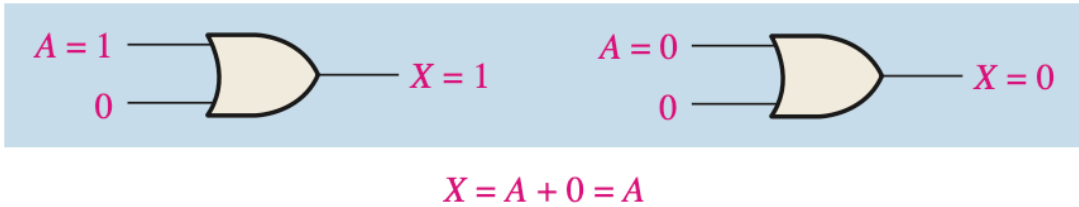


FIGURE 4-8

Rule 2: $A + 1 = 1$ A variable ORed with 1 is always equal to 1. A 1 on an input to an OR gate produces a 1 on the output, regardless of the value of the variable on the other input. This rule is illustrated in Figure 4–9, where the lower input is fixed at 1.

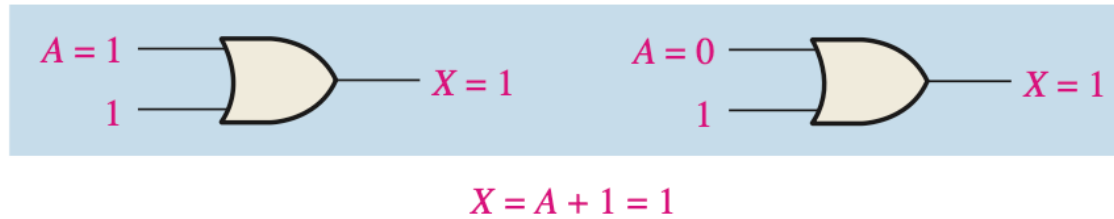


FIGURE 4–9

Rule 3: $A \cdot 0 = 0$ A variable ANDed with 0 is always equal to 0. Any time one input to an AND gate is 0, the output is 0, regardless of the value of the variable on the other input. This rule is illustrated in Figure 4–10, where the lower input is fixed at 0.

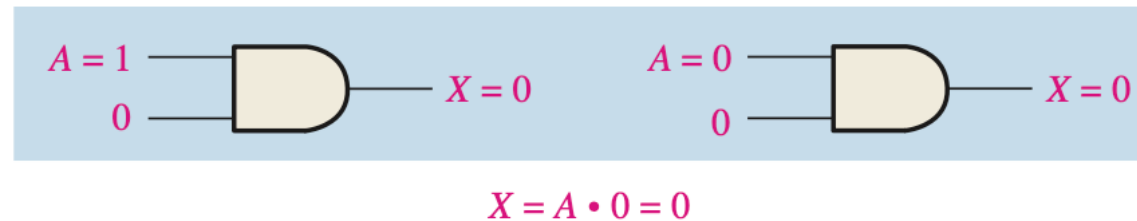


FIGURE 4–10

Rule 4: $A \cdot 1 = A$ A variable ANDed with 1 is always equal to the variable. If A is 0, the output of the AND gate is 0. If A is 1, the output of the AND gate is 1 because both inputs are now 1s. This rule is shown in Figure 4–11, where the lower input is fixed at 1.

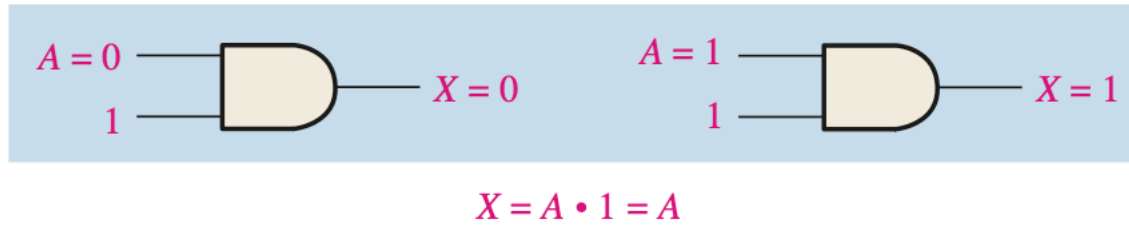


FIGURE 4–11

Rule 5: $A + A = A$ A variable ORed with itself is always equal to the variable. If A is 0, then $0 + 0 = 0$; and if A is 1, then $1 + 1 = 1$. This is shown in Figure 4–12, where both inputs are the same variable.

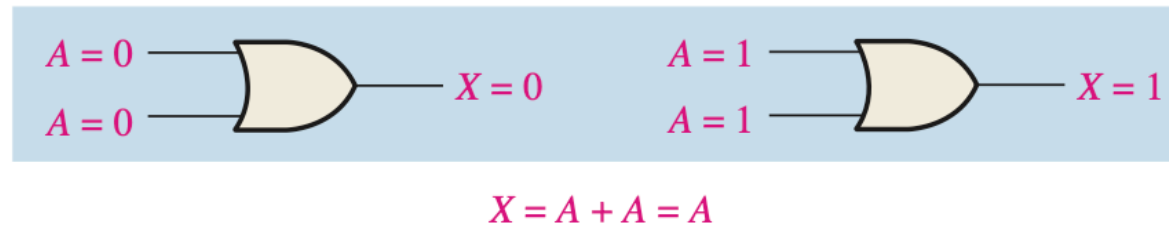


FIGURE 4–12

Rule 6: $A + \bar{A} = 1$ A variable ORed with its complement is always equal to 1. If A is 0, then $0 + \bar{0} = 0 + 1 = 1$. If A is 1, then $1 + \bar{1} = 1 + 0 = 1$. See Figure 4–13, where one input is the complement of the other.

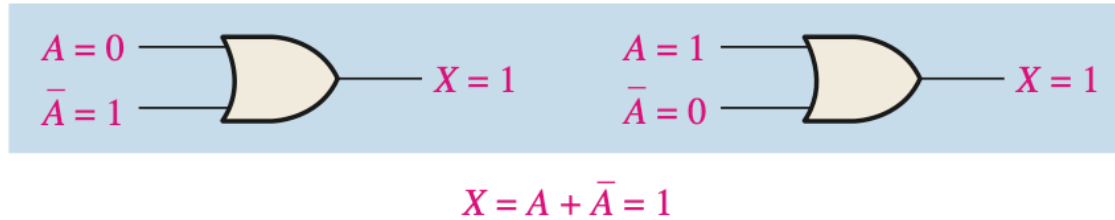


FIGURE 4-13

Rule 7: $A \cdot A = A$ A variable ANDed with itself is always equal to the variable. If $A = 0$, then $0 \cdot 0 = 0$; and if $A = 1$, then $1 \cdot 1 = 1$. Figure 4–14 illustrates this rule.

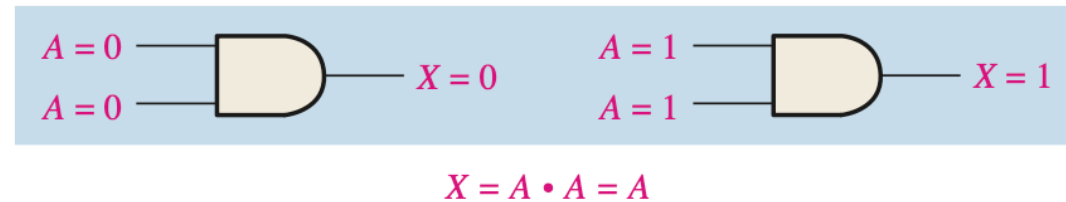


FIGURE 4-14

Rule 8: $A \cdot \bar{A} = 0$ A variable ANDed with its complement is always equal to 0. Either A or $\bar{A}$ will always be 0; and when a 0 is applied to the input of an AND gate, the output will be 0 also. Figure 4–15 illustrates this rule.

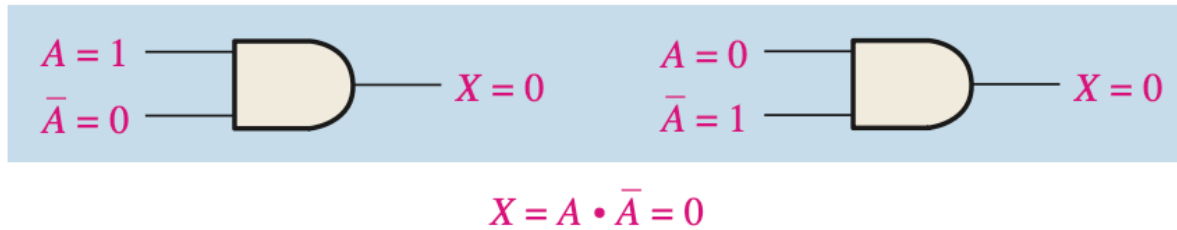


FIGURE 4-15

Rule 9: $\bar{\bar{A}} = A$ The double complement of a variable is always equal to the variable. If you start with the variable A and complement (invert) it once, you get $\bar{A}$. If you then take $\bar{A}$ and complement (invert) it, you get A , which is the original variable. This rule is shown in Figure 4–16 using inverters.

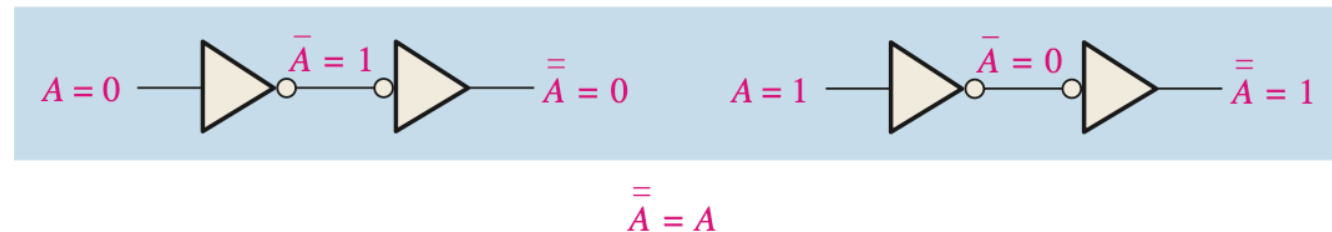


FIGURE 4-16

Rule 10: $A + AB = A$ This rule can be proved by applying the distributive law, rule 2, and rule 4 as follows:

$$\begin{aligned}
 A + AB &= A \cdot 1 + AB = A(1 + B) && \text{Factoring (distributive law)} \\
 &= A \cdot 1 && \text{Rule 2: } (1 + B) = 1 \\
 &= A && \text{Rule 4: } A \cdot 1 = A
 \end{aligned}$$

The proof is shown in Table 4–2, which shows the truth table and the resulting logic circuit simplification.

TABLE 4-2

Rule 10: $A + AB = A$. Open file T04-02 to verify.

A	B	AB	A + AB
0	0	0	0
0	1	0	0
1	0	0	1
1	1	1	1

↑

equal

↑

Rule 11: $A + \bar{A}B = A + B$ This rule can be proved as follows:

$$\begin{aligned}
 A + \bar{A}B &= (A + AB) + \bar{A}B \\
 &= (AA + AB) + \bar{A}B \\
 &= AA + AB + A\bar{A} + \bar{A}B \\
 &= (A + \bar{A})(A + B) \\
 &= 1 \cdot (A + B) \\
 &= A + B
 \end{aligned}$$

Rule 10: $A = A + AB$

Rule 7: $A = AA$

Rule 8: adding $A\bar{A} = 0$

Factoring

Rule 6: $A + \bar{A} = 1$

Rule 4: drop the 1

The proof is shown in Table 4–3, which shows the truth table and the resulting logic circuit simplification.

TABLE 4–3

Rule 11: $A + \bar{A}B = A + B$. Open file T04-03 to verify.

A	B	$\bar{A}B$	$A + \bar{A}B$	$A + B$
0	0	0	0	0
0	1	1	1	1
1	0	0	1	1
1	1	0	1	1

↑

↑

equal

Rule 12: $(A + B)(A + C) = A + BC$ This rule can be proved as follows:

$$\begin{aligned}
 (A + B)(A + C) &= AA + AC + AB + BC && \text{Distributive law} \\
 &= A + AC + AB + BC && \text{Rule 7: } AA = A \\
 &= A(1 + C) + AB + BC && \text{Factoring (distributive law)} \\
 &= A \cdot 1 + AB + BC && \text{Rule 2: } 1 + C = 1 \\
 &= A(1 + B) + BC && \text{Factoring (distributive law)} \\
 &= A \cdot 1 + BC && \text{Rule 2: } 1 + B = 1 \\
 &= A + BC && \text{Rule 4: } A \cdot 1 = A
 \end{aligned}$$

The proof is shown in Table 4–4, which shows the truth table and the resulting logic circuit simplification.

TABLE 4–4

Rule 12: $(A + B)(A + C) = A + BC$. Open file T04-04 to verify.

A	B	C	A + B	A + C	(A + B)(A + C)	BC	A + BC
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	1	1	1	1
1	0	0	1	1	1	0	1
1	0	1	1	1	1	0	1
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1

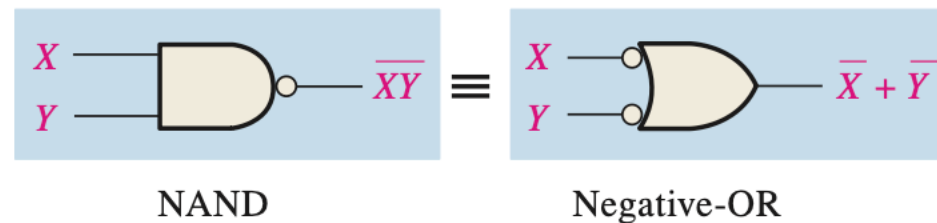
↑ equal ↑

3. DeMorgan's Theorems

DeMorgan's first theorem is stated as follows:

- The complement of a product of variables is equal to the sum of the complements of the variables. **Or**
- The complement of two or more ANDed variables is equivalent to the OR of the complements of the individual variables.
- The formula for expressing this theorem for two variables is:

$$\overline{XY} = \overline{X} + \overline{Y}$$



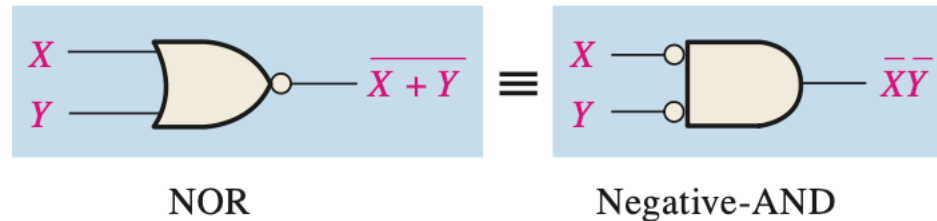
Inputs		Output	
X	Y	$\overline{XY}$	$\overline{X} + \overline{Y}$
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0

3. DeMorgan's Theorems

DeMorgan's second theorem is stated as follows:

- The complement of a sum of variables is equal to the product of the complements of the variables. **Or**
- The complement of two or more ORed variables is equivalent to the AND of the complements of the individual variables.
- The formula for expressing this theorem for two variables is:

$$\overline{X + Y} = \overline{X} \overline{Y}$$



Inputs		Output	
X	Y	$\overline{X + Y}$	$\overline{X} \overline{Y}$
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

Example 4–3

Applying DeMorgan's Theorems

1 Problem

Apply DeMorgan's theorems to the expressions

$$\overline{XYZ} \text{ and } \overline{X + Y + Z}.$$

2 Solution

$$\begin{aligned}\overline{XYZ} &= \bar{X} + \bar{Y} + \bar{Z} \\ \overline{X + Y + Z} &= \bar{X} \bar{Y} \bar{Z}\end{aligned}$$

3 Related Problem

Apply DeMorgan's theorem to the expression

$$\overline{\bar{X} + \bar{Y} + \bar{Z}}.$$

Example 4–4

Applying DeMorgan's Theorems

1 Problem

Apply DeMorgan's theorems to the expressions

$$\overline{WXYZ} \text{ and } \overline{W + X + Y + Z}.$$

2 Solution

$$\begin{aligned}\overline{WXYZ} &= \bar{W} + \bar{X} + \bar{Y} + \bar{Z} \\ \overline{W + X + Y + Z} &= \bar{W} \bar{X} \bar{Y} \bar{Z}\end{aligned}$$

3 Related Problem

Apply DeMorgan's theorem to the expression

$$\overline{\bar{W} \bar{X} \bar{Y} \bar{Z}}.$$

Applying DeMorgan's Theorems

The following procedure illustrates the application of DeMorgan's theorems and Boolean algebra to the specific expression

$$\overline{\overline{A + BC} + D(E + F)}$$

Step 1: Identify the terms to which you can apply DeMorgan's theorems, and think of each term as a single variable. Let $\overline{A + BC} = X$ and $\overline{D(E + F)} = Y$.

Step 2: Since $\overline{X + Y} = \overline{X}\overline{Y}$,

$$\overline{\overline{A + BC} + \overline{D(E + F)}} = \overline{\overline{A + BC}} \overline{\overline{D(E + F)}}$$

Step 3: Use rule 9 ($\overline{\overline{A}} = A$) to cancel the double bars over the left term (this is not part of DeMorgan's theorem).

$$\overline{\overline{A + BC}} \overline{\overline{D(E + F)}} = (A + BC) \overline{\overline{D(E + F)}}$$

Step 4: Apply DeMorgan's theorem to the second term.

$$(A + BC) \overline{\overline{D(E + F)}} = (A + BC) (\overline{\overline{D}} + \overline{\overline{E + F}})$$

Step 5: Use rule 9 ($\overline{\overline{A}} = A$) to cancel the double bars over the $E + \overline{F}$ part of the term.

$$(A + BC) (\overline{\overline{D}} + \overline{\overline{E + \overline{F}}}) = (A + BC) (\overline{\overline{D}} + E + \overline{F})$$

Example 4–5

Applying DeMorgan's Theorems

Apply DeMorgan's theorems to each of the following expressions:

(a) $\overline{(A + B + C)D}$

(b) $\overline{ABC + DEF}$

(c) $\overline{AB + \overline{CD} + EF}$

(a) $\overline{(A + B + C)D}$

Solution

- 1 Let $A + B + C = X$ and $D = Y$.
The expression $\overline{(A + B + C)D}$ is of the form $\overline{XY} = \overline{X} + \overline{Y}$ and can be rewritten as

$$\overline{(A + B + C)D} = \overline{A + B + C} + \overline{D}$$

- 2 Next, apply DeMorgan's theorem to the term $\overline{A + B + C}$.

$$\overline{A + B + C} + \overline{D} = \overline{A}\overline{B}\overline{C} + \overline{D}$$

(b) $\overline{ABC + DEF}$

Solution

- 1 Let $ABC = X$ and $DEF = Y$.
The expression $\overline{ABC + DEF}$ is of the form $\overline{X + Y} = \overline{X}\overline{Y}$ and can be rewritten as

$$\overline{ABC + DEF} = \overline{ABC}\overline{DEF}$$

- 2 Next, apply DeMorgan's theorem to each of the terms $\overline{ABC}$ and $\overline{DEF}$.

$$\overline{ABC}\overline{DEF} = (\overline{A} + \overline{B} + \overline{C})(\overline{D} + \overline{E} + \overline{F})$$

(c) $\overline{AB + \overline{CD} + EF}$

Solution

- 1 Let $\overline{AB} = X$, $\overline{CD} = Y$, and $EF = Z$.
The expression $\overline{AB + \overline{CD} + EF}$ is of the form $\overline{X + Y + Z} = \overline{X}\overline{Y}\overline{Z}$ and can be rewritten as

$$\overline{AB + \overline{CD} + EF} = \overline{AB}\overline{\overline{CD}}\overline{EF}$$

- 2 Next, apply DeMorgan's theorem to each of the terms $\overline{AB}$, $\overline{\overline{CD}}$, and $\overline{EF}$.

$$\overline{AB}\overline{\overline{CD}}\overline{EF} = (\overline{A} + \overline{B})(C + D)(\overline{E} + \overline{F})$$



Related Problem

Apply DeMorgan's theorems to the expression $\overline{ABC + D + E}$.

Example 4–6

Applying DeMorgan's Theorems

Apply DeMorgan's theorems to each expression:

(a) $\overline{(A + B) + \bar{C}}$ (b) $\overline{(A + B) + \bar{C}\bar{D}}$ (c) $\overline{(A + B)\bar{C}\bar{D}} + E + \bar{F}$

Solution

(a) $\overline{(A + B) + \bar{C}} = \overline{\bar{A} + B} \bar{C} = (A + B)C$

(b) $\overline{(A + B) + \bar{C}\bar{D}} = \overline{(A + B)\bar{C}\bar{D}} = (\bar{A} \bar{B})(\bar{C} + \bar{D}) = A\bar{B}(\bar{C} + \bar{D})$

(c) $\overline{(A + B)\bar{C}\bar{D}} + E + \bar{F} = ((\bar{A} + B)\bar{C}\bar{D})(E + \bar{F}) = (\bar{A} \bar{B} + C + D)E\bar{F}$

Related Problem

Apply DeMorgan's theorems to the expression $\overline{AB(C + \bar{D})} + E$.

Example 4-7

Deriving the Exclusive-NOR Expression

1 Problem

The Boolean expression for an exclusive-OR gate is $\overline{A}\overline{B} + \overline{A}B$. With this as a starting point, use DeMorgan's theorems and any other rules or laws that are applicable to develop an expression for the exclusive-NOR gate.

2 Solution

- 1 Complement the XOR expression and apply DeMorgan's theorems.

$$\overline{\overline{A}\overline{B} + \overline{A}B} = (\overline{\overline{A}\overline{B}})(\overline{\overline{A}B}) = (\overline{\overline{A}} + \overline{\overline{B}})(\overline{A} + \overline{B})$$

- 2 Use the distributive law and rule 8 ($A \cdot \overline{A} = 0$).

$$(\overline{\overline{A}} + \overline{\overline{B}})(\overline{A} + \overline{B}) = \overline{\overline{A}}\overline{A} + \overline{\overline{A}}\overline{B} + \overline{\overline{B}}\overline{A} + \overline{\overline{B}}\overline{B} = \overline{\overline{A}}\overline{B} + \overline{\overline{B}}\overline{A}$$

- 3 Final expression for XNOR.

$$\overline{\overline{A}}\overline{B} + \overline{\overline{B}}\overline{A}$$

Equals 1 when both variables are 0 or both are 1.

3 Related Problem

Starting with the expression for a 4-input NAND gate, use DeMorgan's theorems to develop an expression for a 4-input negative-OR gate.

4. Boolean Analysis of Logic Circuits

Boolean Expression for a Logic Circuit

- The expression for the left-most AND gate with inputs C and D is CD .
- The output of the left-most AND gate is one of the inputs to the OR gate and B is the other input. Therefore, the expression for the OR gate is $B + CD$.
- The output of the OR gate is one of the inputs to the right-most AND gate and A is the other input. Therefore, the expression for this AND gate is $A(B + CD)$, which is the final output expression for the entire circuit.

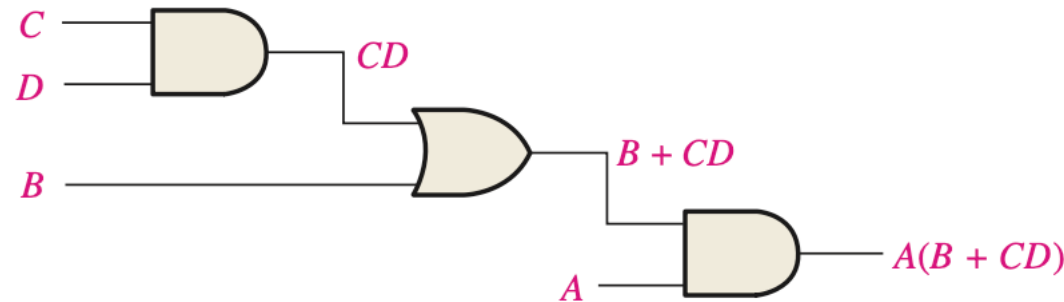


FIGURE 4-18 A combinational logic circuit showing the development of the Boolean expression for the output.

Constructing a Truth Table for a Logic Circuit

Evaluating the expression

- To evaluate the expression $A(B + CD)$, first find the values of the variables that make the expression equal to 1, using the rules for Boolean addition and multiplication.
- The expression $A(B + CD) = 1$ when $A = 1$ and $B = 1$ regardless of the values of C and D or when $A = 1$ and $C = 1$ and $D = 1$ regardless of the value of B . The expression $A(B + CD) = 0$ for all other value combinations of the variables.

Putting the results in truth table Form

- The first step is to list the sixteen input variable combinations of 1s and 0s in a binary sequence.
- Next, place a 1 in the output column for each combination of input variables that was determined in the evaluation.
- Finally, place a 0 in the output column for all other combinations of input variables.

Constructing a Truth Table for a Logic Circuit

TABLE 4-5

Truth table for the logic circuit in Figure 4-18.

Inputs				Output
<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	$A(B + CD)$
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

5. Logic Simplification Using Boolean Algebra

Example 4–9

Simplifying a Boolean Expression

1 Problem

Using Boolean algebra techniques, simplify this expression:

$$AB + A(B + C) + B(B + C)$$

2 Solution

(One possible approach)

Step 1

Distribute the second and third terms.

$$AB + AB + AC + BB + BC$$

Step 2

Use $BB = B$.

$$AB + AB + AC + B + BC$$

Step 3

Use $AB + AB = AB$.

$$AB + AC + B + BC$$

Step 4

Use $B + BC = B$.

$$AB + AC + B$$

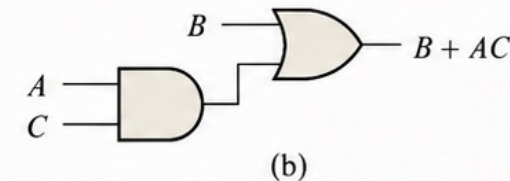
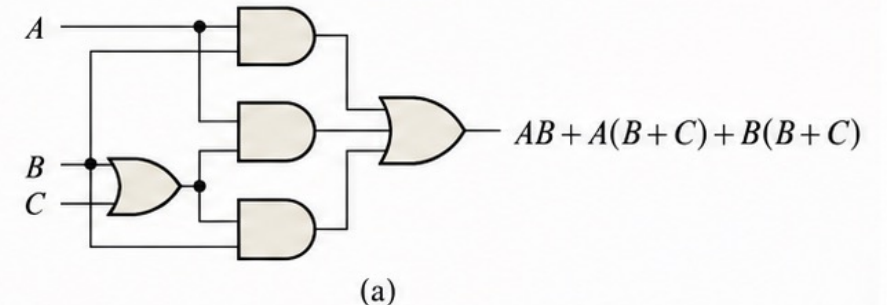
Step 5

Use $AB + B = B$.

$$B + AC$$

3

Equivalent Circuits



These two circuits are equivalent.

4 Related Problem

Simplify the Boolean expression $A\bar{B} + A(\bar{B} + C) + B(\bar{B} + C)$.

Example 4–10

Simplifying a Boolean Expression

1 Problem

Simplify the following Boolean expression:

$$[\overline{A}\overline{B}(C + BD) + \overline{A}\overline{B}]C$$

Brackets mean AND with the term outside.

2 Solution

Step 1

Distribute.

$$(\overline{A}\overline{B}C + \overline{A}\overline{B}BD + \overline{A}\overline{B})C$$

Step 2

Use $\overline{B}B = 0$.

$$(\overline{A}\overline{B}C + A \cdot 0 \cdot D + \overline{A}\overline{B})C$$

Step 3

Use $A \cdot 0 \cdot D = 0$.

$$(\overline{A}\overline{B}C + 0 + \overline{A}\overline{B})C$$

Step 4

Drop the 0.

$$(\overline{A}\overline{B}C + \overline{A}\overline{B})C$$

Step 5

Distribute.

$$\overline{A}\overline{B}CC + \overline{A}\overline{B}C$$

Step 6

Use $CC = C$.

$$\overline{A}\overline{B}C + \overline{A}\overline{B}C$$

Step 7

Factor $\overline{B}C$.

$$\overline{B}C(A + \overline{A})$$

Step 8

Use $A + \overline{A} = 1$.

$$\overline{B}C \cdot 1$$

Step 9

Drop the 1.

$$\overline{B}C$$

3 Related Problem

Simplify the Boolean expression

$$[AB(C + \overline{B}\overline{D}) + \overline{A}\overline{B}]CD.$$

Example 4–11

Simplifying a Boolean Expression

1 Problem

Simplify the following Boolean expression:

$$\bar{A}\bar{B}C + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C}$$

2 Solution

Step 1

Factor BC from the first and last terms.

$$BC(\bar{A} + A) + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C}$$

Step 2

Use $\bar{A} + A = 1$ and factor $A\bar{B}$ from 2nd & 4th.

$$BC \cdot 1 + A\bar{B}(\bar{C} + C) + \bar{A}\bar{B}\bar{C}$$

Step 3

Use $\bar{C} + C = 1$ and drop 1.

$$BC + A\bar{B} \cdot 1 + \bar{A}\bar{B}\bar{C}$$

Step 4

Drop 1.

$$BC + A\bar{B} + \bar{A}\bar{B}\bar{C}$$

Step 5

Factor $\bar{B}$ from 2nd & 3rd.

$$BC + \bar{B}(A + \bar{A}C)$$

Step 6

Use $A + \bar{A}C = A + \bar{C}$.

$$BC + \bar{B}(A + \bar{C})$$

Step 7

Distribute and simplify.

$$BC + A\bar{B} + \bar{B}\bar{C}$$

3 Related Problem

Simplify the Boolean expression

$$A\bar{B}C + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C}$$

Example 4–12

Simplifying a Boolean Expression

1 Problem

Simplify the following Boolean expression:

$$\overline{A}\overline{B} + \overline{A}\overline{C} + \overline{A}BC$$

2 Solution

Step 1 Apply DeMorgan to $\overline{A}\overline{B}$.

$$(\overline{A}\overline{B})(\overline{A}\overline{C}) + \overline{A}BC$$

Step 2 Apply DeMorgan to each term in parentheses.

$$(A + \overline{B})(\overline{A} + \overline{C}) + \overline{A}BC$$

Step 3 Distribute.

$$\overline{A}\overline{A} + \overline{A}\overline{C} + \overline{A}\overline{B} + \overline{B}\overline{C} + \overline{A}BC$$

Step 4 Use $AA = A$ and $\overline{A}\overline{B} + \overline{A}BC = \overline{A}\overline{B}$.

$$\overline{A} + \overline{A}\overline{C} + \overline{A}\overline{B} + \overline{B}\overline{C}$$

Step 5 Use $\overline{A} + \overline{A}\overline{C} = \overline{A}$.

$$\overline{A} + \overline{A}\overline{B} + \overline{B}\overline{C}$$

Step 6 Use $\overline{A} + \overline{A}\overline{B} = \overline{A}$.

$$\overline{A} + \overline{B}\overline{C}$$

3 Related Problem

Simplify the Boolean expression

$$\overline{A}\overline{B} + \overline{A}\overline{C} + \overline{A}BC.$$

6. Standard Forms of Boolean Expressions

The Sum-of-Products (SOP) Form

- When two or more product terms are summed by Boolean addition, the resulting expression is a sum-of-products (SOP).
- An SOP expression can be implemented with one OR gate and two or more AND gates.

$$AB + ABC$$

$$ABC + CDE + \overline{B}C\overline{D}$$

$$\overline{A}B + \overline{A}B\overline{C} + AC$$

Domain of a Boolean Expression

The **domain** of a general Boolean expression is the set of variables contained in the expression in either complemented or uncomplemented form.

AND/OR Implementation of an SOP Expression

- Implementing an SOP expression simply requires ORing the outputs of two or more AND gates.
- A product term is produced by an AND operation, and the sum (addition) of two or more product terms is produced by an OR operation.
- Therefore, an SOP expression can be implemented by AND-OR logic in which the outputs of a number (equal to the number of product terms in the expression) of AND gates connect to the inputs of an OR gate.

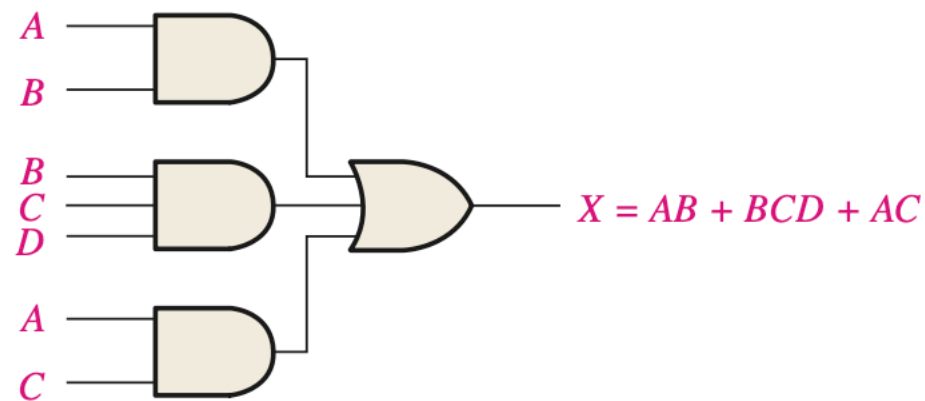


FIGURE 4-22 Implementation of the SOP expression $AB + BCD + AC$.

NAND/NAND Implementation of an SOP Expression

- NAND gates can be used to implement an SOP expression. By using only NAND gates, an AND/OR function can be accomplished, as illustrated in Figure 4–23.
- The first level of NAND gates feed into a NAND gate that acts as a negative-OR gate.
- The NAND and negative-OR inversions cancel and the result is effectively an AND/OR circuit.

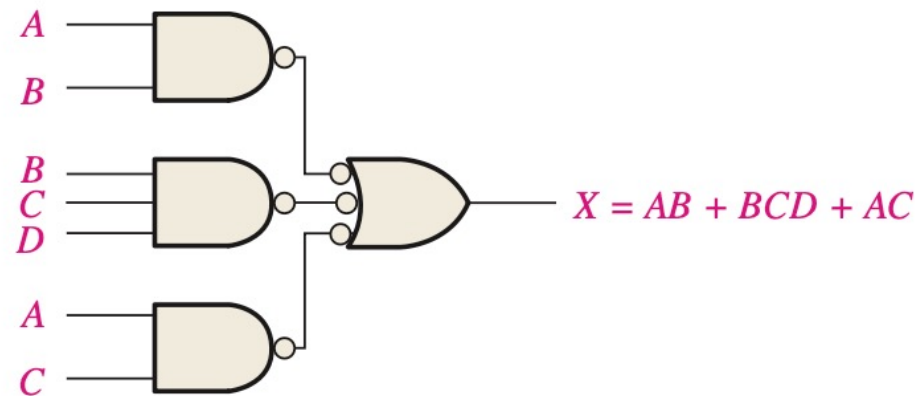


FIGURE 4–23 This NAND/NAND implementation is equivalent to the AND/OR in Figure 4–22.

Conversion of a General Expression to SOP Form

- Any logic expression can be changed into SOP form by applying Boolean algebra techniques.
- For example, the expression $A(B + CD)$ can be converted to SOP form by applying the distributive law: **$A(B + CD) = AB + ACD$**

Example 4–14

Convert to Sum-of-Products Form

1 Problem

Convert each expression to SOP form:

(a) $AB + B(CD + EF)$

(b) $(A + B)(B + C + D)$

(c) $\overline{\overline{(A + B)} + C}$

2 Solution

(a) $AB + B(CD + EF)$

Step 1 Distribute B .

$$AB + BCD + BEF$$

(b) $(A + B)(B + C + D)$

Step 1 Distribute.

$$AB + AC + AD + BB + BC + BD$$

Step 2 Use $BB = B$.

$$AB + AC + AD + B + BC + BD$$

(c) $\overline{\overline{(A + B)} + C}$

Step 1 Apply DeMorgan. $\overline{\overline{(A + B)} + C} = (A + \bar{B})\bar{C}$

Step 2 Distribute. $(A + \bar{B})\bar{C}$

Step 3 Rewrite as SOP. $\bar{A}\bar{C} + \bar{B}\bar{C}$

3 Related Problem

Convert $A\bar{B}\bar{C} + (A + \bar{B})(B + \bar{C} + A\bar{B})$ to SOP form.

The Standard SOP Form

- So far, you have seen SOP expressions in which some of the product terms do not contain all of the variables in the domain of the expression.
- A standard SOP expression is one in which all the variables in the domain appear in each product term in the expression.

Converting Product Terms to Standard Sop

- Step 1: Multiply each nonstandard product term by a term made up of the sum of a missing variable and its complement. This results in two product terms. As you know, you can multiply anything by 1 without changing its value.
- Step 2: Repeat Step 1 until all resulting product terms contain all variables in the domain in either complemented or uncomplemented form. In converting a product term to standard form, the number of product terms is doubled for each missing variable.

Example 4–15

Convert to Standard Sum-of-Products Form

1 Problem

Convert the following Boolean expression to standard SOP form:

$$A\bar{B}C + \bar{A}B + ABCD$$

2 Solution

Step 1 Expand the first term with $D + \bar{D}$.

$$A\bar{B}C = A\bar{B}C(D + \bar{D}) = A\bar{B}CD + A\bar{B}C\bar{D}$$

Step 2 Expand the second term with $C + \bar{C}$ and then $D + \bar{D}$.

$$\begin{aligned}\bar{A}B &= \bar{A}B(C + \bar{C}) = \bar{A}BC + \bar{A}B\bar{C} \\ &= \bar{A}BCD + \bar{A}BC\bar{D} + \bar{A}B\bar{C}D + \bar{A}B\bar{C}\bar{D}\end{aligned}$$

Step 3 The third term, $ABCD$, is already in standard form.

$$ABCD$$

Step 4 Combine all terms.

$$\begin{aligned}A\bar{B}C + \bar{A}B + ABCD &= A\bar{B}CD + A\bar{B}C\bar{D} + \bar{A}BCD + \bar{A}BC\bar{D} \\ &\quad + \bar{A}B\bar{C}D + \bar{A}B\bar{C}\bar{D} + ABCD\end{aligned}$$

3 Related Problem

Convert the expression $\bar{W}\bar{X}Y + \bar{X}Y\bar{Z} + W\bar{X}\bar{Y}$ to standard SOP form.

Binary Representation of a Standard Product Term

- A standard product term is equal to 1 for only one combination of variable values.

$$A\bar{B}C\bar{D} = 1 \cdot \bar{0} \cdot 1 \cdot \bar{0} = 1 \cdot 1 \cdot 1 \cdot 1 = 1$$

- An SOP expression is equal to 1 only if one or more of the product terms in the expression is equal to 1.

Example 4–16

Finding the Binary Values for an SOP Expression

1 Problem

Determine the binary values for which the following standard SOP expression is equal to 1:

$$ABCD + A\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D}$$

2 Solution

Each product term equals 1 for the following values:

Term 1:	$ABCD$	$A = 1, B = 1, C = 1, D = 1$	1111
Term 2:	$A\bar{B}\bar{C}D$	$A = 1, B = 0, C = 0, D = 1$	1001
Term 3:	$\bar{A}\bar{B}\bar{C}\bar{D}$	$A = 0, B = 0, C = 0, D = 0$	0000

The SOP expression equals 1 for any or all of these three combinations.

3 Related Problem

Determine the binary values for which the following SOP expression is equal to 1:

$$\bar{X}YZ + X\bar{Y}Z + XY\bar{Z} + X\bar{Y}\bar{Z} + XYZ$$

Is this a standard SOP expression?

The Product-of-Sums (POS) Form

- When two or more sum terms are multiplied, the resulting expression is a product-of-sums (POS).

$$(\bar{A} + B)(A + \bar{B} + C)$$

$$(\bar{A} + \bar{B} + \bar{C})(C + \bar{D} + E)(\bar{B} + C + D)$$

$$(A + B)(A + \bar{B} + C)(\bar{A} + C)$$

- In a POS expression, a single overbar cannot extend over more than one variable; however, more than one variable in a term can have an overbar.

Implementation of a POS Expression

- Implementing a POS expression simply requires ANDing the outputs of two or more OR gates.
- A sum term is produced by an OR operation, and the product of two or more sum terms is produced by an AND operation.
- Therefore, a POS expression can be implemented by logic in which the outputs of a number (equal to the number of sum terms in the expression) of OR gates connect to the inputs of an AND gate, as Figure 4–24 shows for the expression $(A + B)(B + C + D)(A + C)$. The output X of the AND gate equals the POS expression.

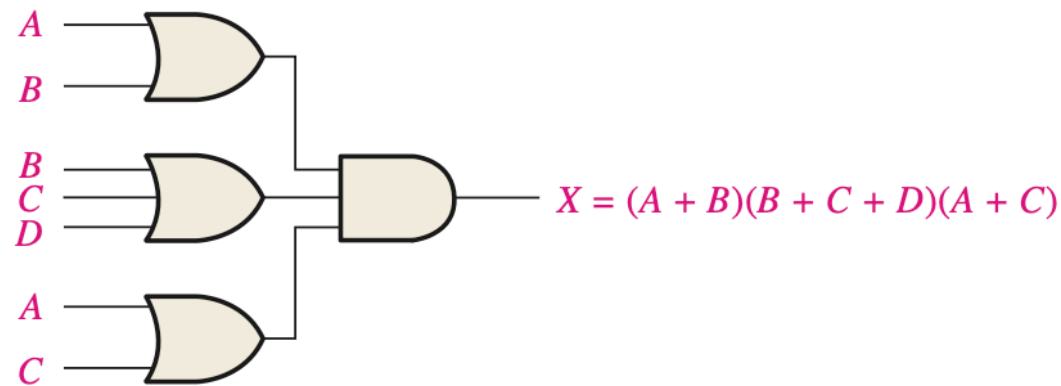


FIGURE 4–24 Implementation of the POS expression $(A + B)(B + C + D)(A + C)$.

The Standard POS Form

- So far, you have seen POS expressions in which some of the sum terms do not contain all of the variables in the domain of the expression.
- A standard POS expression is one in which all the variables in the domain appear in each sum term in the expression. For example, $(\bar{A} + \bar{B} + \bar{C} + \bar{D})(A + \bar{B} + C + D)(A + B + \bar{C} + D)$

Converting a Sum Term to Standard POS

- Step 1: Add to each nonstandard product term a term made up of the product of the missing variable and its complement. This results in two sum terms. As you know, you can add 0 to anything without changing its value.
- Step 2: Apply rule 12 from Table 4–1: $A + BC = (A + B)(A + C)$.
- Step 3: Repeat Step 1 until all resulting sum terms contain all variables in the domain in either complemented or uncomplemented form.

Example 4–17

Convert to Standard Product-of-Sums Form

1 Problem

Convert the following Boolean expression to standard POS form:

$$(A + \bar{B} + C)(\bar{B} + C + \bar{D})(A + \bar{B} + \bar{C} + D)$$

2 Solution

Step 1 Add missing D to the first term.

$$A + \bar{B} + C = (A + \bar{B} + C + D)(A + \bar{B} + C + \bar{D})$$

Step 2 Add missing A to the second term.

$$\bar{B} + C + \bar{D} = (A + \bar{B} + C + \bar{D})(\bar{A} + \bar{B} + C + \bar{D})$$

Step 3 Third term already has all variables.

$$A + \bar{B} + \bar{C} + D$$

Step 4 Write the standard POS form.

$$(A + \bar{B} + C + D)(A + \bar{B} + C + \bar{D})(\bar{A} + \bar{B} + C + \bar{D})(A + \bar{B} + \bar{C} + D)$$

3 Related Problem

Convert $(A + \bar{B})(B + C)$ to standard POS form.

Binary Representation of a Standard Sum Term

- A standard sum term is equal to 0 for only one combination of variable values.

$$A + \overline{B} + C + \overline{D} = 0 + \overline{1} + 0 + \overline{1} = 0 + 0 + 0 + 0 = 0$$

- A POS expression is equal to 0 only if one or more of the sum terms in the expression is equal to 0.

Example 4-18

Finding the Binary Values for a POS Expression

1 Problem

Determine the binary values of the variables for which the following standard POS expression is equal to 0:

$$(A + B + C + D)(A + \bar{B} + \bar{C} + D)(\bar{A} + \bar{B} + \bar{C} + \bar{D})$$

2 Solution

Each sum term equals 0 for the following values:

Term 1:	$A + B + C + D$	$A = 0, B = 0, C = 0, D = 0$	0000
Term 2:	$A + \bar{B} + \bar{C} + D$	$A = 0, B = 1, C = 1, D = 0$	0110
Term 3:	$\bar{A} + \bar{B} + \bar{C} + \bar{D}$	$A = 1, B = 1, C = 1, D = 1$	1111

The POS expression equals 0 when any of the three sum terms equals 0.

3 Related Problem

Determine the binary values for which the following POS expression is equal to 0:

$$(X + \bar{Y} + Z)(\bar{X} + Y + Z)(X + Y + \bar{Z})(\bar{X} + \bar{Y} + \bar{Z})(X + \bar{Y} + \bar{Z})$$

Is this a standard POS expression?

Converting Standard SOP to Standard POS

The binary values of the product terms in a given standard SOP expression are not present in the equivalent standard POS expression. Also, the binary values that are not represented in the SOP expression are present in the equivalent POS expression.

- Step 1: Evaluate each product term in the SOP expression. That is, determine the binary numbers that represent the product terms.
- Step 2: Determine all of the binary numbers not included in the evaluation in Step 1.
- Step 3: Write the equivalent sum term for each binary number from Step 2 and express in POS form.

Using a similar procedure, you can go from POS to SOP.

EXAMPLE 4–19

Convert the following SOP expression to an equivalent POS expression:

$$\overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}BC + A\overline{B}C + A\overline{B}\overline{C}$$

Solution

Step 1 List the minterms (binary values) in the SOP.

$$000 + 010 + 011 + 101 + 111$$

Step 2 Find the missing combinations (where the function is 0).

Missing: 001, 100, 110

Step 3 Write a maxterm for each missing combination.

$$001 \rightarrow (A + B + \overline{C})$$

$$100 \rightarrow (\overline{A} + B + C)$$

$$110 \rightarrow (\overline{A} + \overline{B} + C)$$

Step 4 Form the POS expression (product of the maxterms).

$$(A + B + \overline{C})(\overline{A} + B + C)(\overline{A} + \overline{B} + C)$$

<i>A</i>	<i>B</i>	<i>C</i>	In SOP?	In POS?
0	0	0	1	–
0	0	1	0	✓
0	1	0	1	–
0	1	1	1	–
1	0	0	0	✓
1	0	1	1	–
1	1	0	0	✓
1	1	1	1	–

The POS expression contains the combinations that make the function 0.

7. Boolean Expressions and Truth Tables

Converting SOP Expressions to Truth Table Format

- The first step in constructing a truth table is to list all possible combinations of binary values of the variables in the expression.
- Next, convert the SOP expression to standard form if it is not already.
- Finally, place a 1 in the output column (X) for each binary value that makes the standard SOP expression a 1 and place a 0 for all the remaining binary values.

EXAMPLE 4–20

Develop a truth table for the standard SOP expression $\bar{A}BC + A\bar{B}\bar{C} + ABC$.

Solution

Inputs			Output	
<i>A</i>	<i>B</i>	<i>C</i>	<i>X</i>	Product Term
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	1	$\bar{A}BC$
1	0	0	1	$A\bar{B}\bar{C}$
1	0	1	0	
1	1	0	0	
1	1	1	1	ABC

$\bar{A}BC$: 011; $A\bar{B}\bar{C}$: 100; ABC : 111. Place a 1 in the output column for these combinations and 0 for the others.

Related Problem

Create a truth table for the standard SOP expression $\bar{A}\bar{B}\bar{C} + A\bar{B}C$.

Converting POS Expressions to Truth Table Format

- To construct a truth table from a POS expression, list all the possible combinations of binary values of the variables just as was done for the SOP expression.
- Next, convert the POS expression to standard form if it is not already.
- Finally, place a 0 in the output column (X) for each binary value that makes the expression a 0 and place a 1 for all the remaining binary values. .

EXAMPLE 4–21

Developing a Truth Table for a POS Expression

1 Problem

Determine the truth table for the following standard POS expression:

$$(A + B + C)(A + \bar{B} + C)(A + \bar{B} + \bar{C})(\bar{A} + B + \bar{C})(\bar{A} + \bar{B} + \bar{C})$$

2 Solution

- Three variables $\rightarrow$ 8 combinations
- The expression is 0 for: 000, 010, 011, 101, 110
- Place 0 for these rows, and 1 for the others.

3 Related Problem

Develop a truth table for the following standard POS expression:

$$(A + \bar{B} + C)(A + B + \bar{C})(\bar{A} + \bar{B} + \bar{C})$$

TABLE 4–7

Inputs			Output	
A	B	C	X	Sum Term
0	0	0	0	$(A + B + C)$
0	0	1	1	
0	1	0	0	$(A + \bar{B} + C)$
0	1	1	0	$(A + \bar{B} + \bar{C})$
1	0	0	1	
1	0	1	0	$(\bar{A} + B + \bar{C})$
1	1	0	0	$(\bar{A} + \bar{B} + C)$
1	1	1	1	

Note: This truth table is the same as in Example 4–20, so the SOP expression and this POS expression are equivalent.

Determining Standard Expressions from a Truth Table

- To determine the standard SOP expression represented by a truth table, list the binary values of the input variables for which the output is 1. Convert each binary value to the corresponding product term by replacing each 1 with the corresponding variable and each 0 with the corresponding variable complement.

$$1010 \longrightarrow A\bar{B}C\bar{D}$$

- To determine the standard POS expression represented by a truth table, list the binary values for which the output is 0. Convert each binary value to the corresponding sum term by replacing each 1 with the corresponding variable complement and each 0 with the corresponding variable.

$$1001 \longrightarrow \bar{A} + B + C + \bar{D}$$

EXAMPLE 4-22

Finding Standard SOP and POS from a Truth Table

From the truth table in Table 4-8, determine the standard SOP expression and the equivalent standard POS expression.

TABLE 4-8

Inputs			Output
A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

1 Standard SOP Expression

The output is 1 for 011, 100, 110, and 111. Convert to product terms:

$$011 \rightarrow \bar{A}BC$$

$$100 \rightarrow A\bar{B}\bar{C}$$

$$110 \rightarrow AB\bar{C}$$

$$111 \rightarrow ABC$$

Standard SOP expression:

$$X = \bar{A}BC + A\bar{B}\bar{C} + AB\bar{C} + ABC$$

2 Standard POS Expression

The output is 0 for 000, 001, 010, and 101. Convert to sum terms:

$$000 \rightarrow A + B + C$$

$$001 \rightarrow A + B + \bar{C}$$

$$010 \rightarrow A + \bar{B} + C$$

$$101 \rightarrow \bar{A} + B + \bar{C}$$

Standard POS expression:

$$X = (A + B + C)(A + B + \bar{C})(A + \bar{B} + C)(\bar{A} + B + \bar{C})$$

8. The Karnaugh Map

- A Karnaugh map (K-Map) provides a systematic method for simplifying Boolean expressions and, if properly used, will produce the simplest SOP or POS expression possible, known as the minimum expression.
- A Karnaugh map is similar to a truth table because it presents all of the possible values of input variables and the resulting output for each value.
- The number of cells in a Karnaugh map, as well as the number of rows in a truth table, is equal to the total number of possible input variable combinations. For three variables, the number of cells is $2^3 = 8$. For four variables, the number of cells is $2^4 = 16$.

The purpose of a Karnaugh map is to simplify a Boolean expression.

The 3-Variable Karnaugh Map

- The 3-variable Karnaugh map is an array of eight cells, as shown in Figure 4–25(a).
- Binary values of A and B are along the left side (notice the sequence) and the values of C are across the top. The value of a given cell is the binary values of A and B at the left in the same row combined with the value of C at the top in the same column.
- Figure 4–25(b) shows the standard product terms that are represented by each cell in the Karnaugh map.

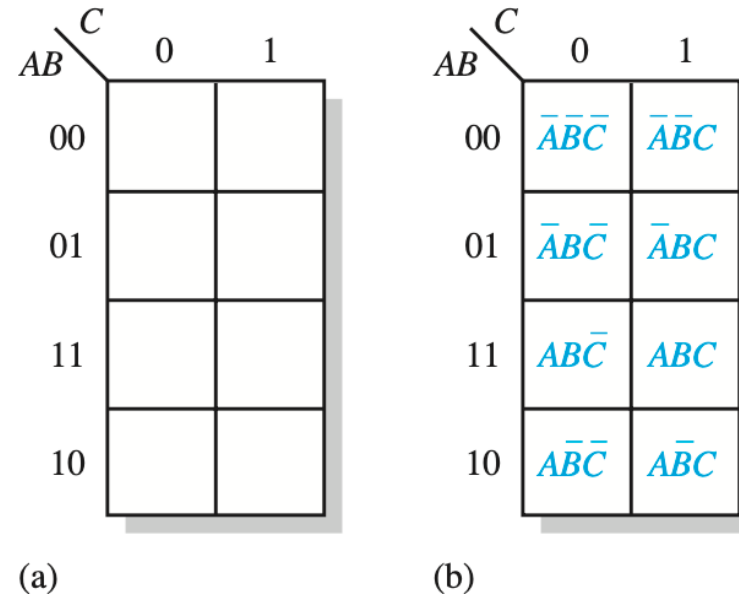


FIGURE 4–25 A 3-variable Karnaugh map showing Boolean product terms for each cell.

The 4-Variable Karnaugh Map

- The 4-variable Karnaugh map is an array of sixteen cells, as shown in Figure 4–26(a).
- Binary values of A and B are along the left side and the values of C and D are across the top. The value of a given cell is the binary values of A and B at the left in the same row combined with the binary values of C and D at the top in the same column.
- Figure 4–26(b) shows the standard product terms that are represented by each cell in the 4-variable Karnaugh map.

$AB \backslash CD$					
		00	01	11	10
AB	00				
	01				
	11				
	10				

(a)

$AB \backslash CD$					
		00	01	11	10
AB	00	$\bar{A}\bar{B}\bar{C}\bar{D}$	$\bar{A}\bar{B}\bar{C}D$	$\bar{A}\bar{B}C\bar{D}$	$\bar{A}\bar{B}CD$
	01	$\bar{A}B\bar{C}\bar{D}$	$\bar{A}B\bar{C}D$	$\bar{A}BC\bar{D}$	$\bar{A}BCD$
	11	$AB\bar{C}\bar{D}$	$AB\bar{C}D$	$ABC\bar{D}$	$ABCD$
	10	$A\bar{B}\bar{C}\bar{D}$	$A\bar{B}\bar{C}D$	$A\bar{B}C\bar{D}$	$A\bar{B}CD$

(b)

FIGURE 4–26 A 4-variable Karnaugh map.

Cell Adjacency

- The cells in a Karnaugh map are arranged so that there is only a single-variable change between adjacent cells. *Adjacency is defined by a single-variable change.*
- Physically, each cell is adjacent to the cells that are immediately next to it on any of its four sides. A cell is not adjacent to the cells that diagonally touch any of its corners.
- Also, the cells in the top row are adjacent to the corresponding cells in the bottom row and the cells in the outer left column are adjacent to the corresponding cells in the outer right column. This is called “wrap-around” adjacency.

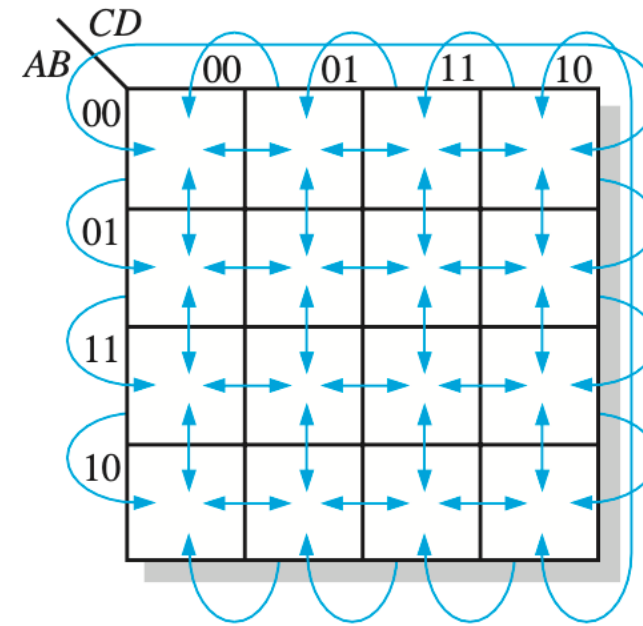


FIGURE 4-27 Adjacent cells on a Karnaugh map are those that differ by only one variable. Arrows point between adjacent cells.

9. Karnaugh Map SOP Minimization

Mapping a Standard SOP Expression

- For an SOP expression in standard form, a 1 is placed on the Karnaugh map for each product term in the expression. Each 1 is placed in a cell corresponding to the value of a product term.
- Step 1: Determine the binary value of each product term in the standard SOP expression. After some practice, you can usually do the evaluation of terms mentally.
- Step 2: As each product term is evaluated, place a 1 on the Karnaugh map in the cell having the same value as the product term.

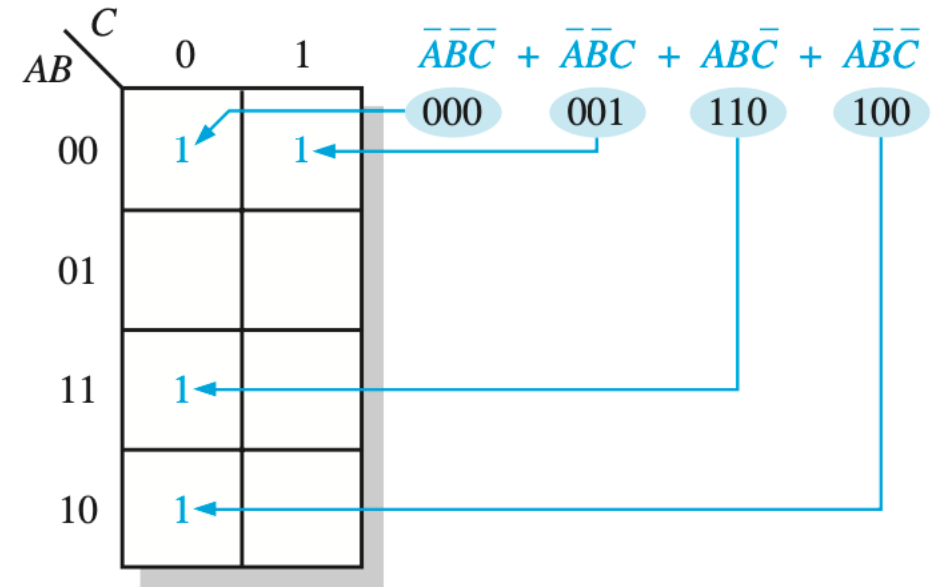


FIGURE 4-28 Example of mapping a standard SOP expression.

EXAMPLE 4-23

Mapping a Standard SOP Expression on a Karnaugh Map

1 Problem

Map the following standard SOP expression on a Karnaugh map:

$$\bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$$

2 Solution

Place a 1 on the 3-variable Karnaugh map for each product term in the expression.

$$\begin{array}{cccc} \bar{A}\bar{B}\bar{C} & + & \bar{A}\bar{B}C & + & A\bar{B}\bar{C} & + & ABC \\ 001 & & 010 & & 110 & & 111 \end{array}$$

3 Related Problem

Map the standard SOP expression $\bar{A}BC + A\bar{B}C + A\bar{B}\bar{C}$ on a Karnaugh map.

3-Variable Karnaugh Map

$\begin{array}{c} C \\ AB \end{array}$		0	1	
00			1	$\bar{A}\bar{B}C$
01		1		$\bar{A}B\bar{C}$
11		1	1	ABC
10				$AB\bar{C}$

EXAMPLE 4-24

Mapping a Standard SOP Expression on a Karnaugh Map

1 Problem

Map the following standard SOP expression on a Karnaugh map:

$$\bar{A}BCD + \bar{A}\bar{B}\bar{C}D + AB\bar{C}D + ABCD + AB\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D}$$

2 Solution

Evaluate the expression as shown below. Place a 1 on the 4-variable Karnaugh map for each product term in the expression.

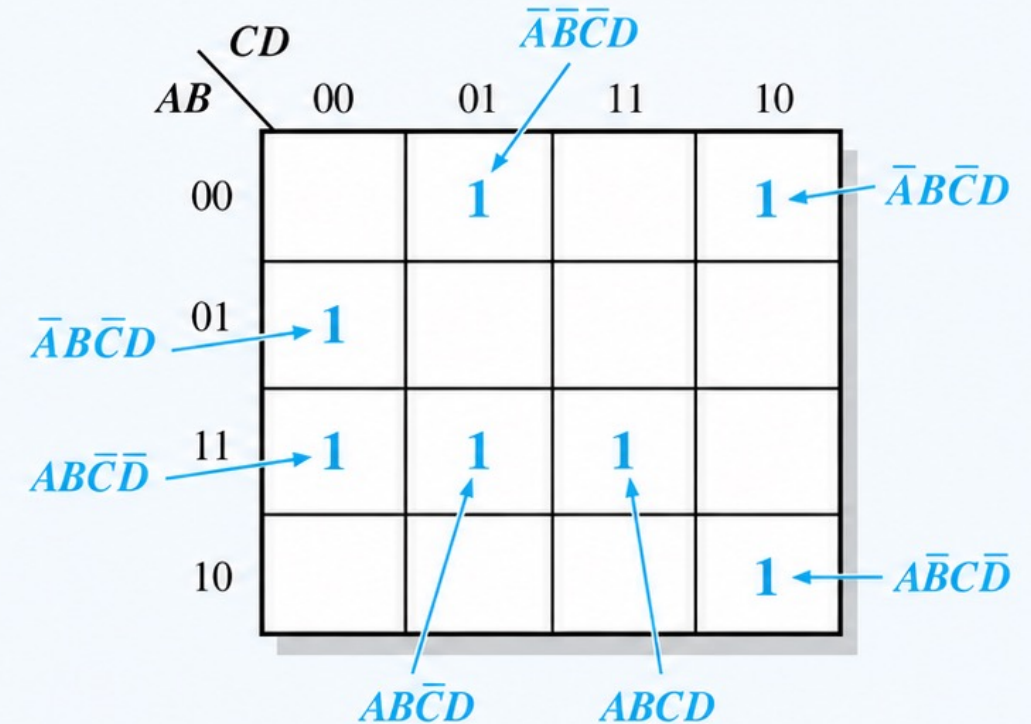
$$\begin{array}{ccccccc} \bar{A}BCD & + & \bar{A}\bar{B}\bar{C}D & + & AB\bar{C}D & + & ABCD & + & AB\bar{C}\bar{D} & + & \bar{A}\bar{B}\bar{C}\bar{D} & + & \bar{A}\bar{B}C\bar{D} \\ 0011 & & 0100 & & 1101 & & 1111 & & 1100 & & 0001 & & 1010 \end{array}$$

3 Related Problem

Map the following standard SOP expression on a Karnaugh map:

$$ABCD + ABC\bar{D} + AB\bar{C}D + ABCD$$

4-Variable Karnaugh Map



EXAMPLE 4–25

Mapping a Nonstandard SOP Expression on a Karnaugh Map

1 Problem

Map the following SOP expression on a Karnaugh map:

$$\overline{A} + A\overline{B} + ABC\overline{C}.$$

2 Solution

- The expression is not in standard form.
- First expand the terms numerically:

$\overline{A}$	$A\overline{B}$	$ABC\overline{C}$
000	100	110
001	101	
010		
011		

- Map each of the resulting binary values by placing a 1 in the appropriate cell of the 3-variable Karnaugh map.

3-Variable Karnaugh Map

		C	
		0	1
AB	00	1	1
	01	1	1
	11	1	
	10	1	1

3 Related Problem

Map the SOP expression $BC + \overline{A}\overline{C}$ on a Karnaugh map.

EXAMPLE 4-26

Mapping a Nonstandard SOP Expression on a Karnaugh Map

1 Problem

Map the following SOP expression on a Karnaugh map:

$$\overline{B}\overline{C} + A\overline{B} + AB\overline{C} + A\overline{B}C\overline{D} + \overline{A}B\overline{C}D + A\overline{B}CD$$

2 Solution

- The expression is not in standard form.
- Expand the terms by including all combinations of the missing variables:

$\overline{B}\overline{C}$	$A\overline{B}$	$AB\overline{C}$	$A\overline{B}C\overline{D}$	$\overline{A}B\overline{C}D$	$A\overline{B}CD$
0000	0100	1010	1010	0001	1011
0001	0101	1011			
1000	1100				
1001	1101				

- Map each of the resulting binary values by placing a 1 in the appropriate cell of the 4-variable Karnaugh map.
- Some of the values in the expanded expression are redundant.

4-Variable Karnaugh Map

$AB \backslash CD$	00	01	11	10
00	1	1		
01				
11	1	1		
10	1	1	1	1

3 Related Problem

Map the SOP expression $BC + A\overline{C}$ on a Karnaugh map.

Karnaugh Map Simplification of SOP Expressions

Grouping the 1s

You can group 1s on the Karnaugh map according to the following rules by enclosing those adjacent cells containing 1s. The goal is to maximize the size of the groups and to minimize the number of groups.

- 1** A group must contain either 1, 2, 4, 8, or 16 cells, which are all powers of two. In the case of a 3-variable map, $2^3 = 8$ cells is the maximum group.
- 2** Each cell in a group must be adjacent to one or more cells in that same group, but all cells in the group do not have to be adjacent to each other.
- 3** Always include the largest possible number of 1s in a group in accordance with rule 1.
- 4** Each 1 on the map must be included in at least one group. The 1s already in a group can be included in another group as long as the overlapping groups include noncommon 1s.

Example Groups on a 3-Variable K-Map

		C	
		0	1
AB	00	1	1
	01		1
	11		1
	10	1	1

Groups can be different sizes and can overlap.



Goal: Use the largest possible groups to minimize the number of groups.

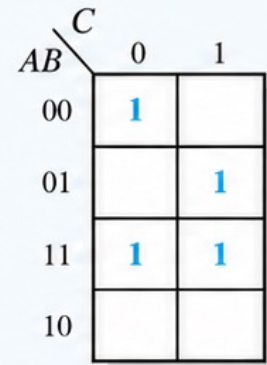
EXAMPLE 4-27

Grouping 1s on Karnaugh Maps

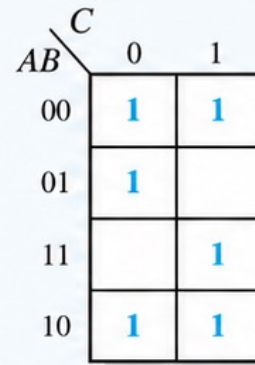
1 Problem

Group the 1s in each of the Karnaugh maps in Figure 4-33.

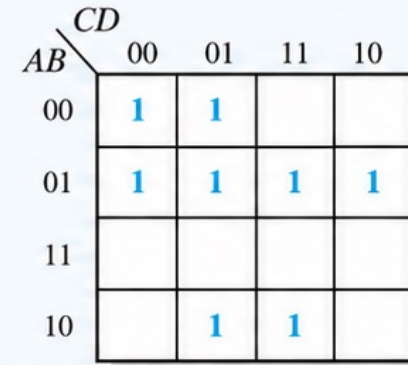
FIGURE 4-33 Given Karnaugh maps



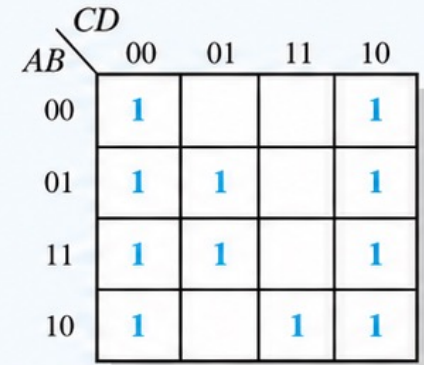
(a)



(b)



(c)

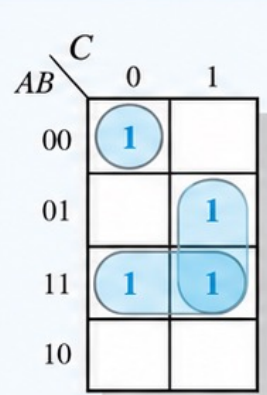


(d)

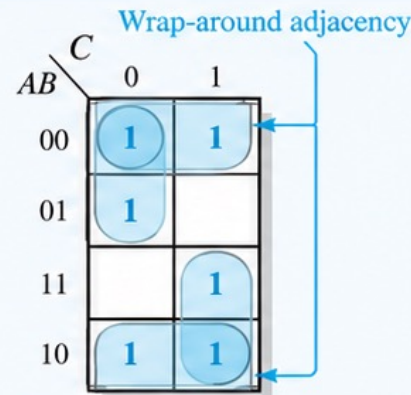
2 Solution

The groupings are shown in Figure 4-34. In some cases, there may be more than one way to group the 1s to form maximum groupings.

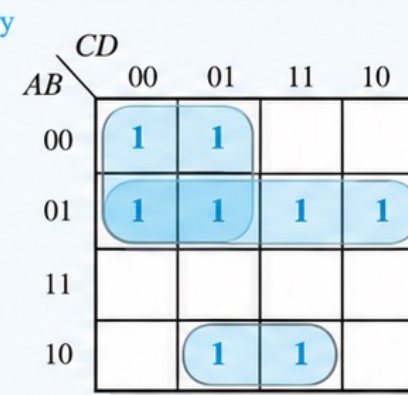
FIGURE 4-34 Grouped Karnaugh maps



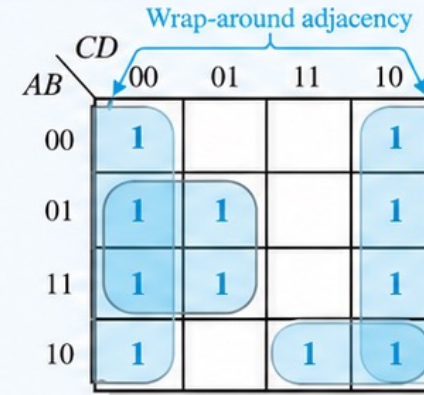
(a)



(b)



(c)



(d)

Determining the Minimum SOP Expression from the Map

After all the 1s are properly mapped and grouped, apply the following rules to find the minimum product terms and the minimum SOP expression.

1 Form groups of 1s

Each group of adjacent 1s creates one product term.

2 Keep only consistent variables

In each group, include only variables that appear in the same form (all 0s or all 1s). Eliminate variables that appear in both forms (contradictory variables).

3 Determine the product term for each group

The product term contains only the consistent variables from the group.

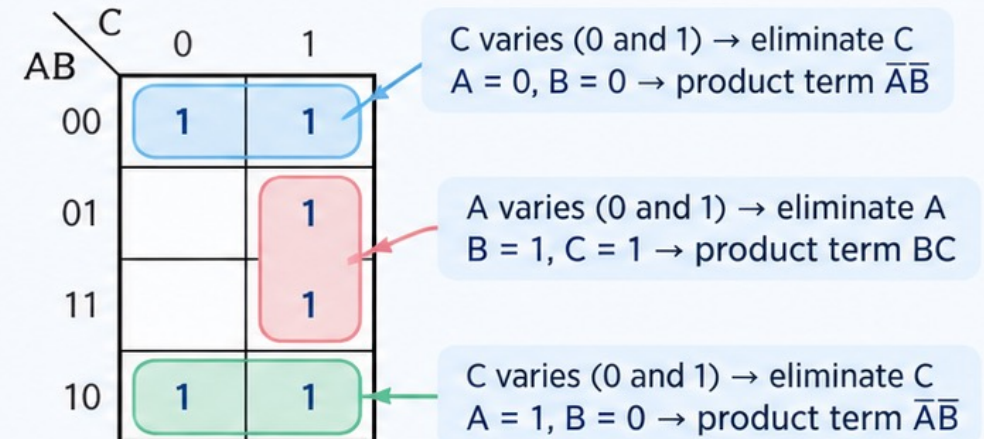
4 Sum the product terms

Add all product terms obtained from the groups to get the minimum SOP expression.



Goal: Find the minimum SOP expression.

Example: Forming a Product Term



Group Size (3 Variables)

- 1 1 cell → 3-variable term
- 2 2 cells → 2-variable term
- 3 4 cells → 1-variable term
- 4 8 cells → 1 (constant)

Group Size (4 Variables)

- 1 1 cell → 4-variable term
- 2 2 cells → 3-variable term
- 3 4 cells → 2-variable term
- 4 8 cells → 1-variable term
- 5 16 cells → 1 (constant)

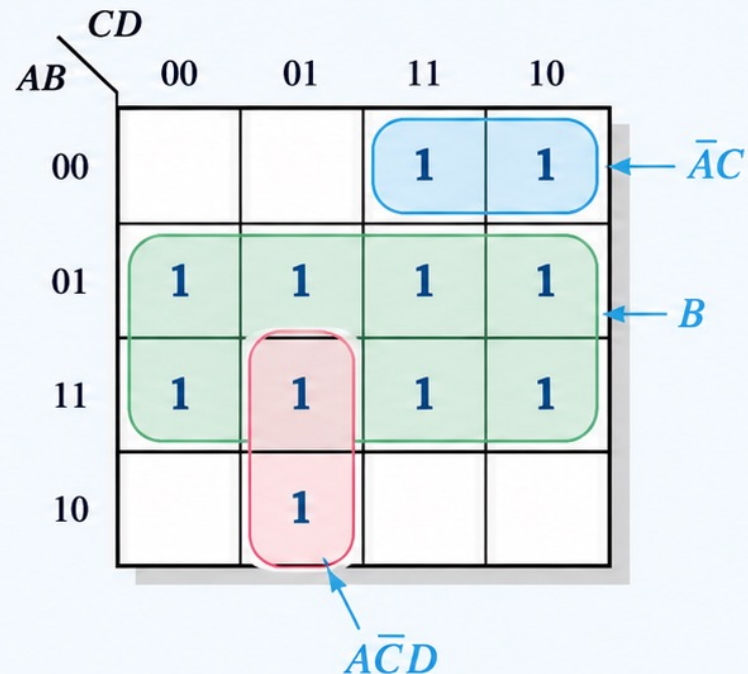
EXAMPLE 4–28

Determining the Minimum SOP Expression from a Karnaugh Map

1 Problem

Determine the product terms for the Karnaugh map in Figure 4–35 and write the resulting minimum SOP expression.

Karnaugh Map (Figure 4–35)



2 Solution

- 8-cell group $\rightarrow B$**
Cells contain both A and $\bar{A}$, C and $\bar{C}$, D and $\bar{D} \rightarrow$ eliminate A, C, D . Product term: B .
- 4-cell group $\rightarrow \bar{A}C$**
Cells contain $B, \bar{B}, D$, and $\bar{D} \rightarrow$ eliminate B and D . Product term: $\bar{A}C$.
- 2-cell group $\rightarrow A\bar{C}D$**
Cells contain B and $\bar{B} \rightarrow$ eliminate B . Product term: $A\bar{C}D$.

Minimum SOP Expression

$$F = B + \bar{A}C + A\bar{C}D$$

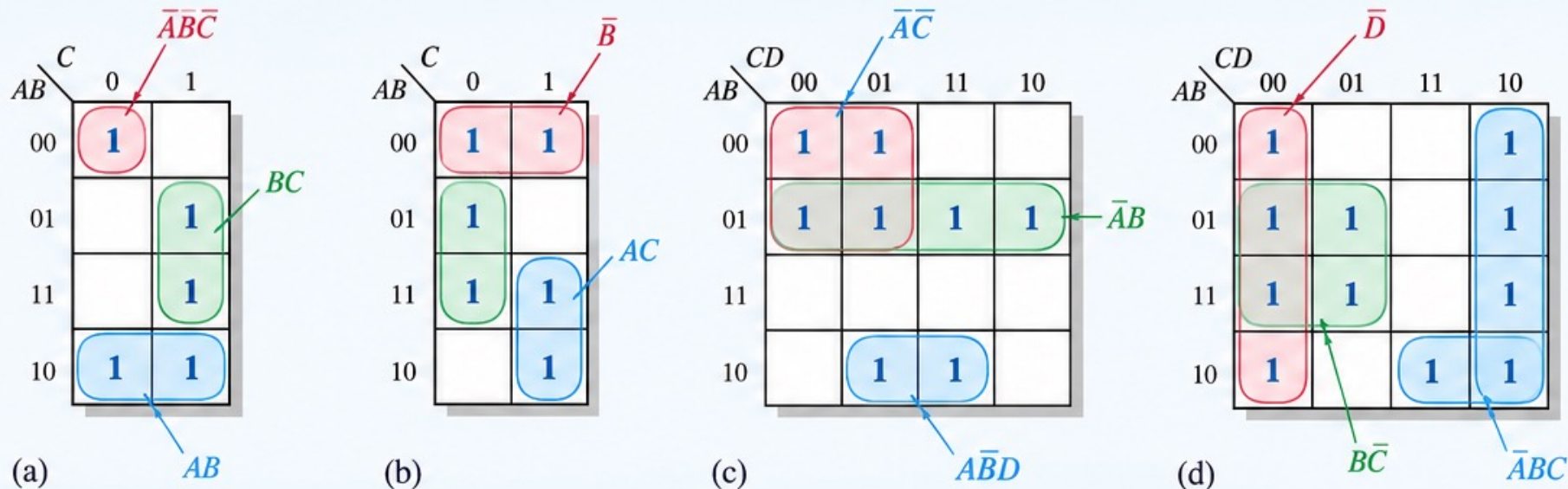
EXAMPLE 4–29

Determining the Minimum SOP Expression from Karnaugh Maps

1 Problem

Determine the product terms for each of the Karnaugh maps in Figure 4–36 and write the resulting minimum SOP expression.

FIGURE 4–36 Given Karnaugh maps



2 Solution

Form the product term for each group by keeping only the variables that are the same within the group and sum the terms to get the minimum SOP expression.

(a) 3-Variable K-Map

- 1 1-cell group $\rightarrow \bar{A}\bar{B}\bar{C}$
- 2 2-cell group $\rightarrow BC$
- 3 2-cell group $\rightarrow AB$

Minimum SOP expression:

$$F = AB + BC + \bar{A}\bar{B}\bar{C}$$

(b) 3-Variable K-Map

- 1 2-cell group $\rightarrow \bar{B}$
- 2 2-cell group $\rightarrow \bar{A}\bar{C}$
- 3 2-cell group $\rightarrow AC$

Minimum SOP expression:

$$F = \bar{B} + \bar{A}\bar{C} + AC$$

(c) 4-Variable K-Map

- 1 4-cell group $\rightarrow \bar{A}\bar{C}$
- 2 4-cell group $\rightarrow \bar{A}\bar{B}$
- 3 2-cell group $\rightarrow \bar{A}\bar{B}\bar{D}$

Minimum SOP expression:

$$F = \bar{A}\bar{B} + \bar{A}\bar{C} + \bar{A}\bar{B}\bar{D}$$

(d) 4-Variable K-Map

- 1 4-cell group $\rightarrow \bar{D}$
- 2 4-cell group $\rightarrow \bar{B}\bar{C}$
- 3 4-cell group $\rightarrow \bar{A}\bar{B}\bar{C}$

Minimum SOP expression:

$$F = \bar{D} + \bar{A}\bar{B}\bar{C} + \bar{B}\bar{C}$$

EXAMPLE 4–30

Minimizing a Standard SOP Expression Using a Karnaugh Map

1 Problem

Use a Karnaugh map to minimize the following standard SOP expression:

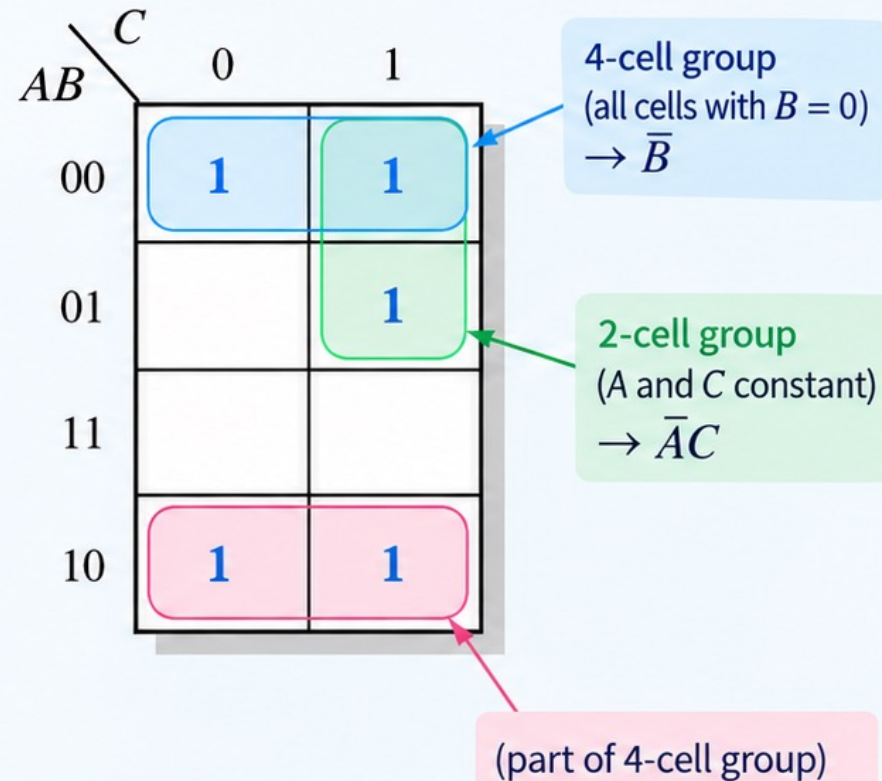
$$\begin{aligned} &A\bar{B}C + \bar{A}BC + \bar{A}\bar{B}C \\ &+ AB\bar{C} + A\bar{B}\bar{C} \end{aligned}$$

2 Minterms

The binary values are
 $101 + 011 + 001 + 000 + 100$

Place 1s on the 3-variable Karnaugh map.

3 Karnaugh Map and Groupings



4 Solution

1 4-cell group (top and bottom rows)

Within the group, $B = 0$ is constant.
Product term: $\bar{B}$

2 2-cell group (right column, top two cells)

$A = 0$ and $C = 1$ are constant.
Product term: $\bar{A}C$

Minimum SOP Expression

$$F = \bar{B} + \bar{A}C$$

EXAMPLE 4–31

Minimizing an SOP Expression Using a Karnaugh Map

1 Problem

Use a Karnaugh map to minimize the following SOP expression:

$$\bar{B}\bar{C}D + A\bar{B}C\bar{D} + AB\bar{C}\bar{D} + \bar{A}BCD \\ + A\bar{B}CD + \bar{A}B\bar{C}D + ABC\bar{D} + AB\bar{C}\bar{D}$$

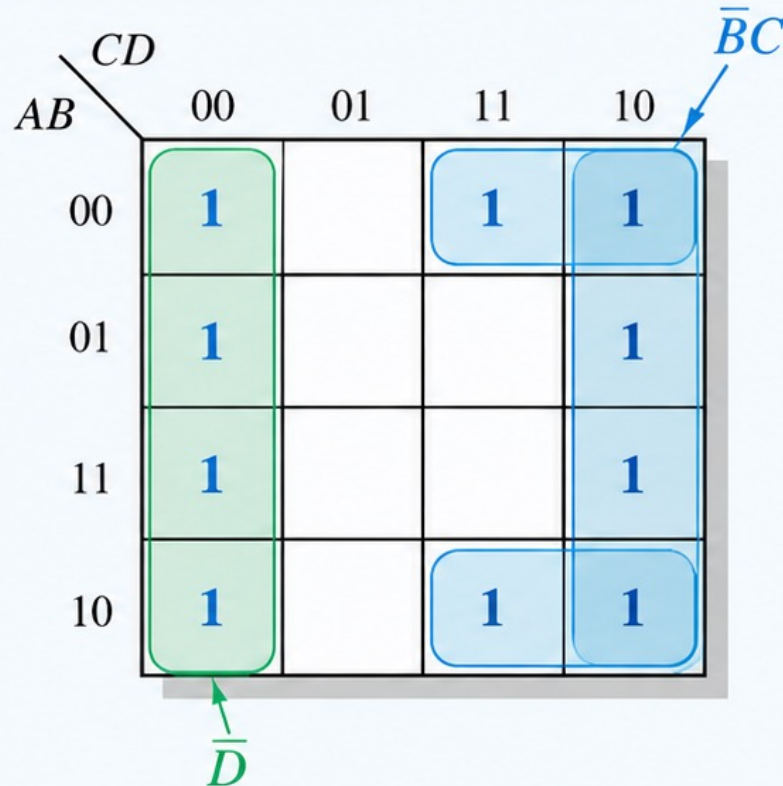
2 Make terms standard

The first term $\bar{B}\bar{C}D$ is missing A , so expand it:

$$\bar{B}\bar{C}D = A\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}D$$

Map all terms on the 4-variable Karnaugh map.

3 Karnaugh Map and Groupings



4 Solution

- 8-cell group (left column)**
Cells in the left column have $D = 0$ (constant), others change.
Product term: $\bar{D}$
- 4-cell group (right column)**
Cells in the right column have $B = 0$ and $C = 1$ (constant), others change.
Product term: $\bar{B}C$

Minimum SOP Expression

$$F = \bar{D} + \bar{B}C$$

Mapping Directly from a Truth Table

$$X = \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + A\overline{B}\overline{C} + ABC$$

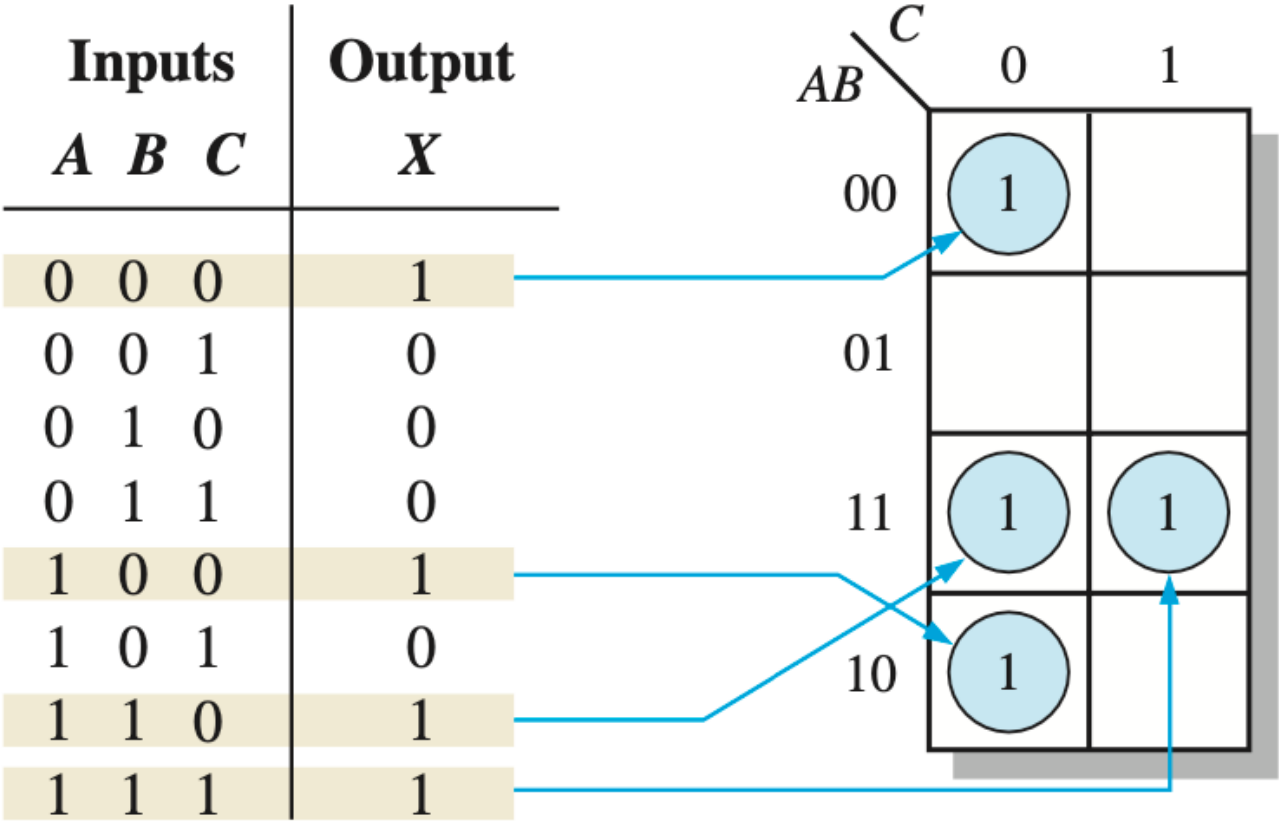
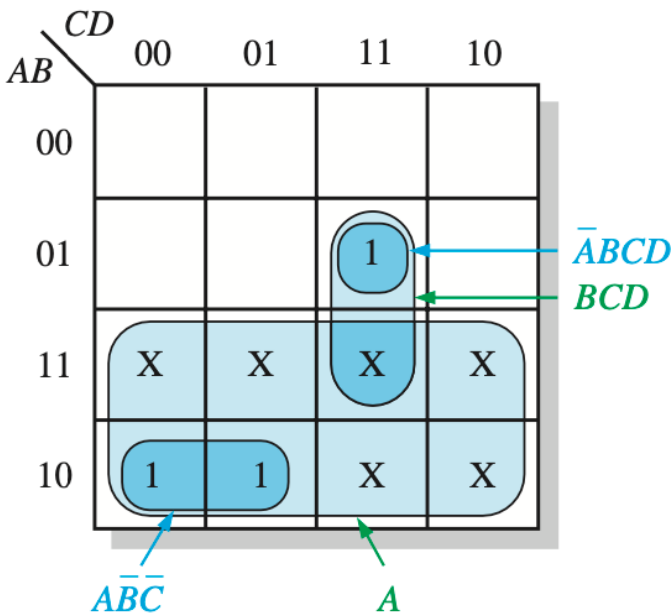


FIGURE 4–39 Example of mapping directly from a truth table to a Karnaugh map.

“Don’t Care” Conditions

Inputs				Output
A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Don't cares



(a) Truth table

(b) Without “don’t cares” $Y = \bar{A}\bar{B}\bar{C} + \bar{A}BCD$
With “don’t cares” $Y = A + BCD$

FIGURE 4–40 Example of the use of “don’t care” conditions to simplify an expression.

EXAMPLE 4-32

7-Segment Display: Minimizing the Expression for Segment a

1 Problem

In a 7-segment display, segment a is activated for 0, 2, 3, 5, 6, 7, 8, and 9. Represent each digit with a BCD code, write the SOP expression for segment a, and minimize it using a Karnaugh map.

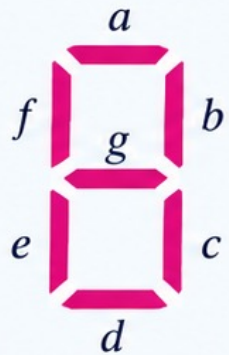


FIGURE 4-41

7-segment display.

2 Standard SOP Expression

Segment a is ON for digits 0, 2, 3, 5, 6, 7, 8, 9:

Digits: 0 2 3 5 6 7 8 9
BCD: 0000 0010 0011 0101 0110 0111 1000 1001

Standard SOP expression:

$$a = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}BC\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}C\bar{D}$$

X's (don't cares): 1010, 1011, 1100, 1101, 1110, 1111
(these BCD codes do not occur for decimal digits).

3 Karnaugh Map and Groupings

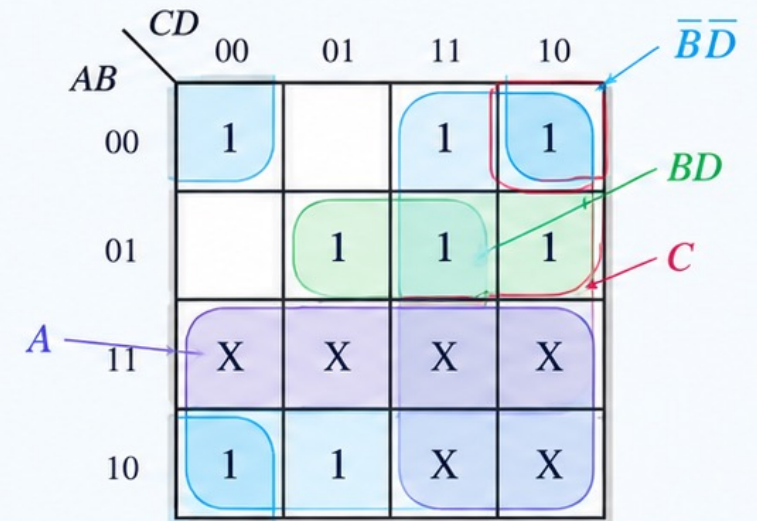


FIGURE 4-42

4 Solution

From the Karnaugh map, the groups yield the product terms:

- 1 Group (corners: 4 cells) $\rightarrow A$
- 2 Group (middle row: 4 cells) $\rightarrow C$
- 3 Group (top-right 2 cells) $\rightarrow BD$
- 4 Group (top-left 2 cells) $\rightarrow \bar{B}\bar{D}$

Minimized Expressions

$$a = A + C + BD + \bar{B}\bar{D}$$

10. Karnaugh Map POS Minimization

Mapping a Standard POS Expression

- For a POS expression in standard form, a 0 is placed on the Karnaugh map for each sum term in the expression. Each 0 is placed in a cell corresponding to the value of a sum term.
- Step 1: Determine the binary value of each sum term in the standard POS expression. This is the binary value that makes the term equal to 0.
- Step 2: As each sum term is evaluated, place a 0 on the Karnaugh map in the corresponding cell.

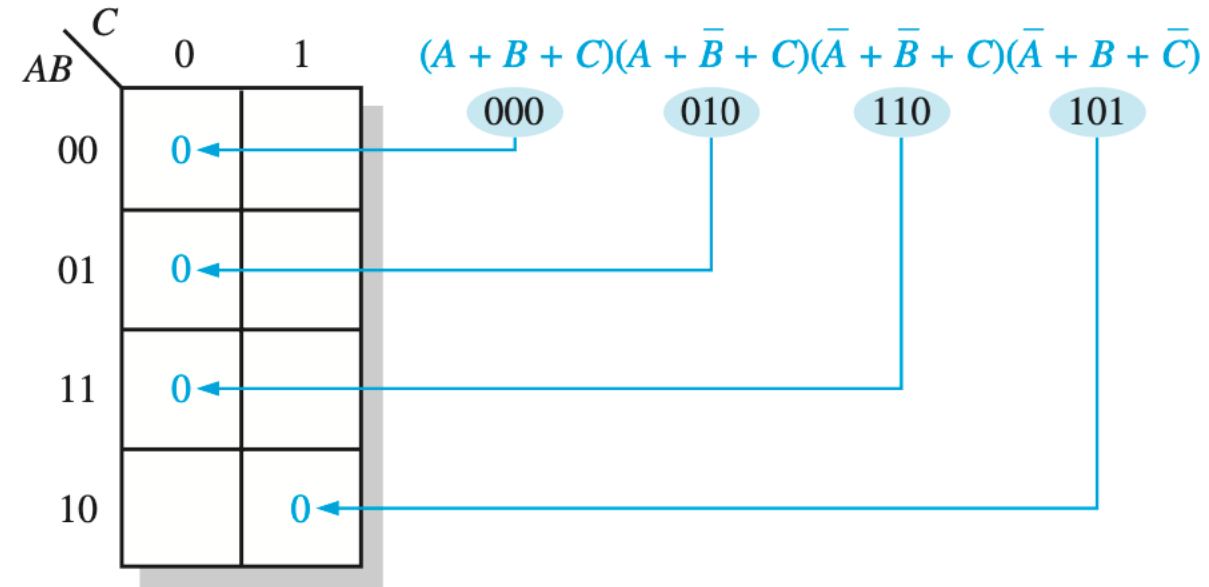


FIGURE 4-43 Example of mapping a standard POS expression.

EXAMPLE 4–33

Mapping a Standard POS Expression on a Karnaugh Map

1 Problem

Map the following standard POS expression on a K-map:

$$\begin{aligned} &(\bar{A} + \bar{B} + C + D)(A + B + \bar{C} + \bar{D}) \\ &(A + B + \bar{C} + D)(\bar{A} + \bar{B} + C + \bar{D}) \\ &\quad (A + B + \bar{C} + D) \end{aligned}$$

2 Evaluate the terms

A POS term is 0 only when all its literals are 0. Find the 4-bit code for each term:

$$\begin{aligned} (\bar{A} + \bar{B} + C + D) &\rightarrow 1100 \\ (A + B + \bar{C} + \bar{D}) &\rightarrow 1011 \\ (A + B + \bar{C} + D) &\rightarrow 0010 \\ (\bar{A} + \bar{B} + C + \bar{D}) &\rightarrow 1111 \\ (A + B + \bar{C} + D) &\rightarrow 0011 \end{aligned}$$

3 Karnaugh Map

Place a 0 in each cell corresponding to the codes: 1100, 1011, 0010, 1111, 0011.

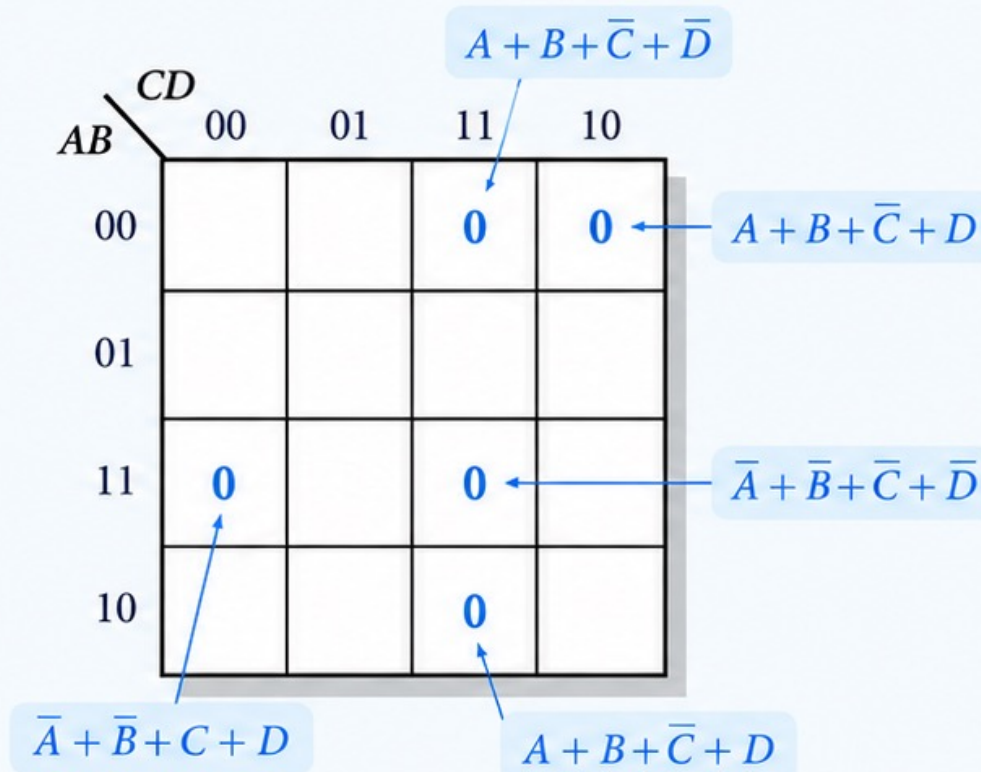


FIGURE 4–44

4 Result

The 4-variable K-map with 0s for the POS term is shown in Figure 4–44.



Key Point:

- In a POS expression, each term corresponds to a single 0 on the K-map.
- A term $(X_1 + X_2 + X_3 + X_4)$ is 0 when all literals are 0.



Related Problem

Map the following standard POS expression on a K-map:

$$\begin{aligned} &(A + \bar{B} + C + D)(A + B + \bar{C} + \bar{D}) \\ &(A + B + C + D)(\bar{A} + B + \bar{C} + \bar{D}) \end{aligned}$$

EXAMPLE 4–34

Minimizing a Standard POS Expression (and Equivalent SOP)

1 Problem

Use a Karnaugh map to minimize the following standard POS expression and also derive the equivalent SOP expression.

$$(A + B + C)(A + B + \bar{C})$$

$$(A + \bar{B} + C)(A + \bar{B} + \bar{C})$$

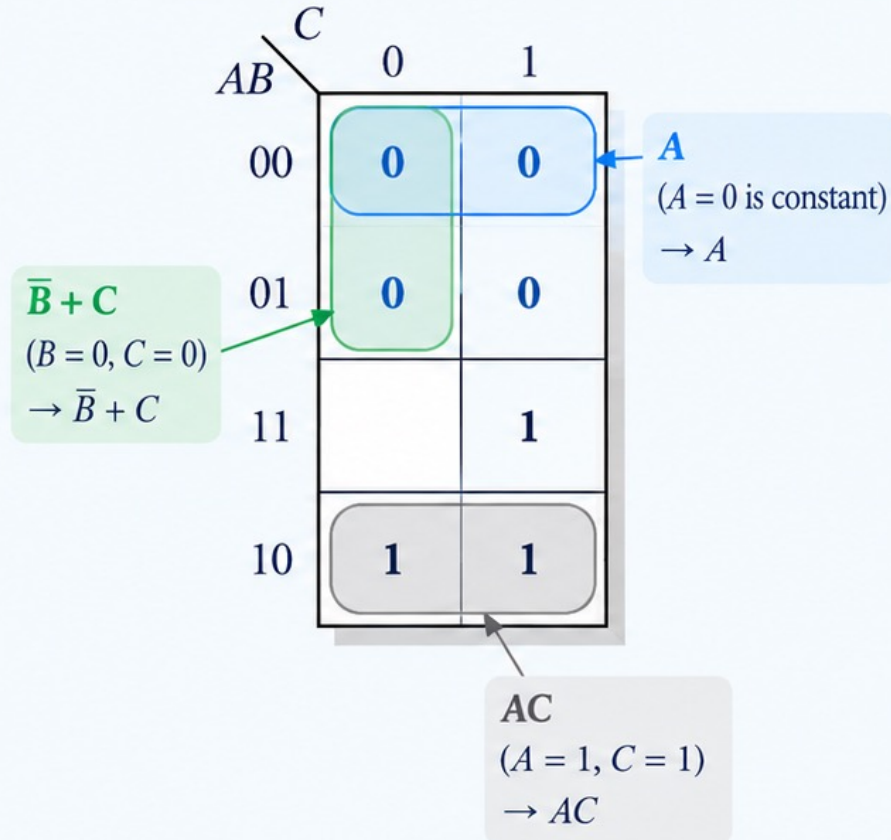
2 Find the 0s

A sum term is 0 when all literals are 0. The combinations of binary values are

$$\begin{array}{cccc} 000 & 001 & 010 & 011 \\ (0 + 0 + 0) & (0 + 0 + 1) & (0 + 1 + 0) & (0 + 1 + 1) \end{array}$$

Place 0s on the 3-variable Karnaugh map.

3 K-map and Groupings



4 Result

Minimum POS expression:

$$(A)(\bar{B} + C) = A(\bar{B} + C)$$

Equivalent SOP expression:

$$AC + A\bar{B} = A(\bar{B} + C)$$



Key Point:

- For a POS expression, place 0s on the map and group adjacent 0s.
- Each group of 0s yields one sum term.
- Grouping the 1s (gray) gives an equivalent SOP expression.

EXAMPLE 4-35

Minimizing a Standard POS Expression Using a Karnaugh Map

1 Problem

Use a Karnaugh map to minimize the following POS expression:

$$(B + C + D)(A + B + \bar{C} + D) \\ (\bar{A} + B + C + \bar{D})(A + \bar{B} + C + D) \\ (\bar{A} + \bar{B} + C + D)$$

2 Make it standard

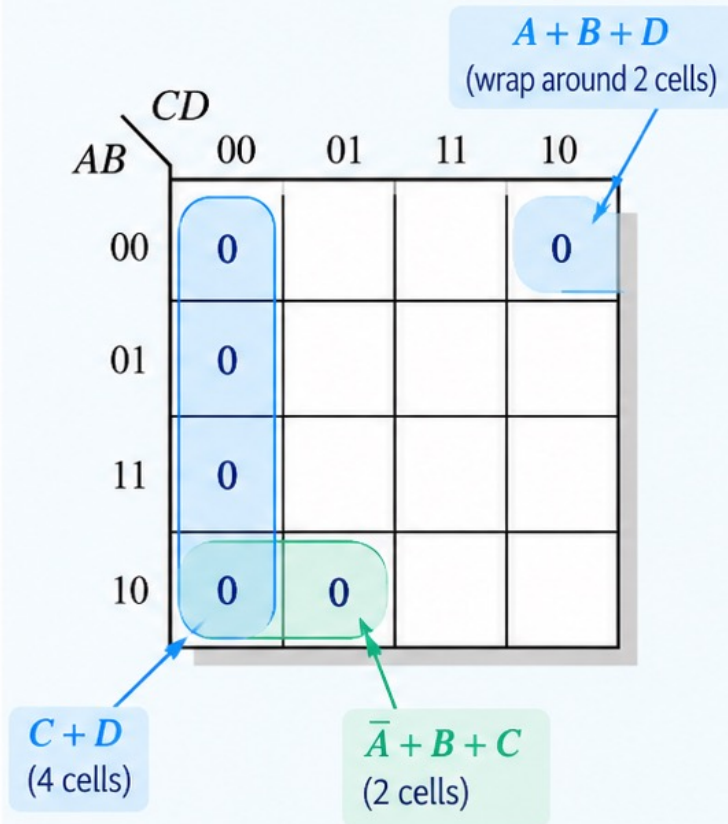
Expand the first term (missing A):

$$(B + C + D) = (\bar{A} + B + C + D)(A + B + C + D)$$

Standard POS expression:

$$(\bar{A} + B + C + D) \\ (A + B + C + D) \\ (A + B + \bar{C} + D) \\ (\bar{A} + B + C + \bar{D}) \\ (A + \bar{B} + C + D) \\ (\bar{A} + \bar{B} + C + D)$$

3 Karnaugh Map and Groupings



4 Result

The groups yield the product terms:

- 1 Left column (4 cells) $\rightarrow C + D$
- 2 Bottom-left pair (2 cells) $\rightarrow \bar{A} + B + C$
- 3 Top corners (2 cells) $\rightarrow A + B + D$

Minimum POS Expression

$$(C + D)(\bar{A} + B + C)(A + B + D)$$



Key Point

- In a POS expression, place 0s on the K-map and group adjacent 0s.
- Each group of 0s gives one sum term.
- Use wrap around adjacency (edges are adjacent).
- This minimized expression is equivalent to the original expression.

EXAMPLE 4–36

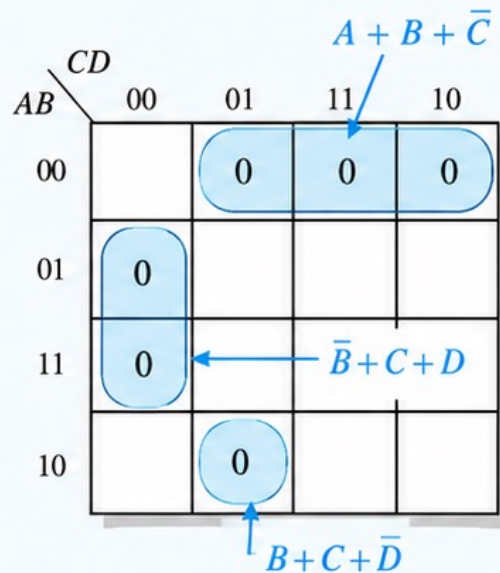
From POS to Minimum POS, Standard SOP, and Minimum SOP

1 Problem

Using a K-map, convert the following standard POS expression to a minimum POS expression, a standard SOP expression, and a minimum SOP expression.

$$(\bar{A} + \bar{B} + C + D)(A + \bar{B} + C + D) \\ (A + B + C + \bar{D})(A + B + C + \bar{D}) \\ (\bar{A} + B + C + \bar{D})(A + B + \bar{C} + D)$$

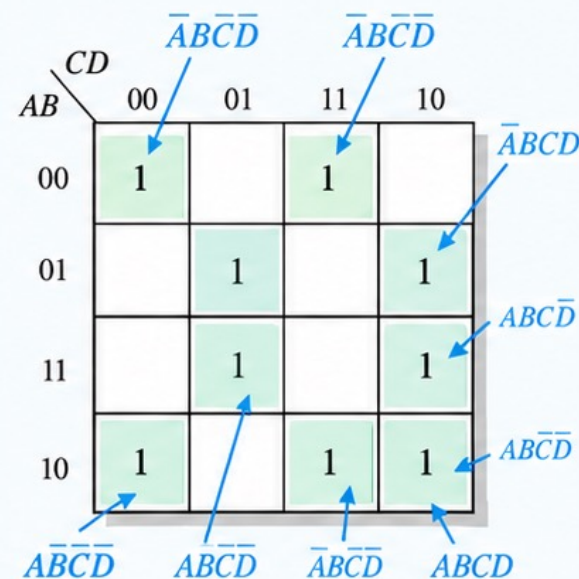
2 Minimum POS (group 0s)



Minimum POS expression

$$(A + B + \bar{C})(\bar{B} + C + D)(B + C + \bar{D})$$

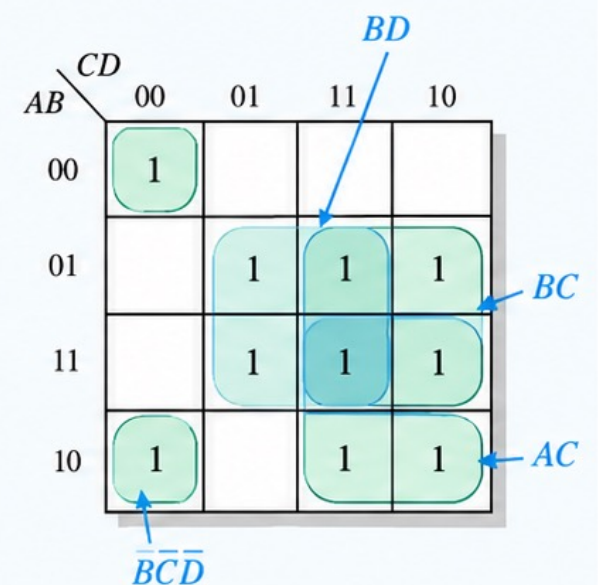
3 Standard SOP (from 1s)



Standard SOP expression

$$\bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}BC\bar{D} + \bar{A}BCD + AB\bar{C}\bar{D} + AB\bar{C}D + ABC\bar{D} + ABCD$$

4 Minimum SOP (group 1s)



Minimum SOP expression

$$AC + BC + BD + \bar{B}\bar{C}\bar{D}$$



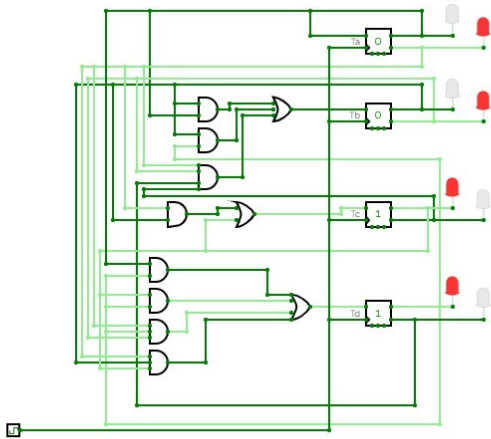
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THE END

Lecture 4: Boolean Algebra and Logic Simplification



INSTRUCTOR: Dr. Vuong Quoc Bao